

## Final note (Fourier Analysis)

09.09.2023

Q-1: Define signal, Stationary Signal, DC Signal.

⇒ Signal: Signal is the physical quantity that is measurable. Signals are functions of time. System is a physical entity that exists. Signal is produced from a system.

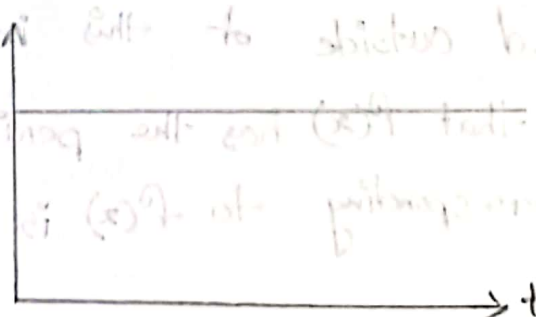
Stationary Signal: Stationary Signals are further divided into deterministic and random Signals.

Random Signals are unpredictable in their frequency content and their amplitude level, but they still have relatively uniform statistical characteristics over time.

DC Signal: In electronic circuits things happen. Voltage/time,  $V/t$ , graphs provide a useful method of describing

the changes which take place.

The diagram below shows the  $V/t$  graph, which represents a DC Signal.



Direct current (DC) is produced by sources such as batteries, thermocouples, solar cells etc.

Q-2: What is periodic signal? Write the characteristics of periodic signal.

$\Rightarrow$  A signal which is repeating itself is a periodic signal.

Periodic signal characteristics:-

- i. Amplitude (A): Signal value, measured in volts.
- ii. Frequency (f): repetition rate, cycles per second or Hertz.
- iii. Period (T): Amount of time it takes for one repetition
- iv. Phase ( $\phi$ ): Relative position in time, measured in degrees.

Q-3: Define Fourier series in the interval  $(-L, L)$

Sol<sup>n</sup>: let  $f(x)$  be defined in the interval  $(-L, L)$  and determined outside of this interval by  $f(x+2L) = f(x)$  i.e. assume that  $f(x)$  has the period  $2L$ . The Fourier expansion corresponding to  $f(x)$  is defined by



$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Where the Fourier coefficients  $a_n$  and  $b_n$  are given by

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{2L} dx$$

$$\text{and } b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{2L} dx$$

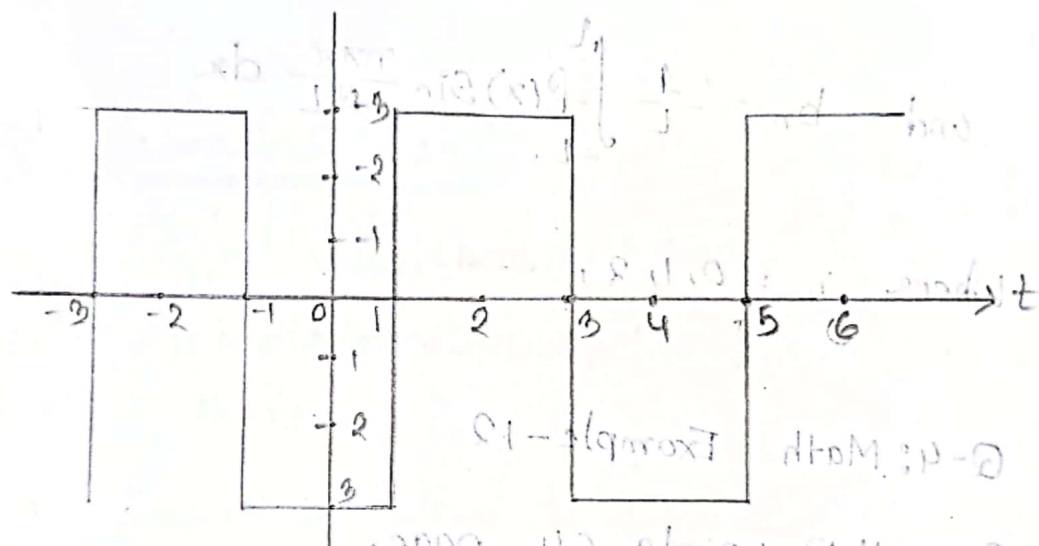
Where  $n = 0, 1, 2, \dots$

Q-4: Math Example-12.

Q-5: H.W up to 64 page.

Example 6:  $f(t) = -3 \quad ; \quad -1 \leq t \leq 1$   
 $= 3 \quad ; \quad 1 \leq t < 3$

$-f(t) = -f(t+4)$  [This period is 4]



Period =  $P = 4$

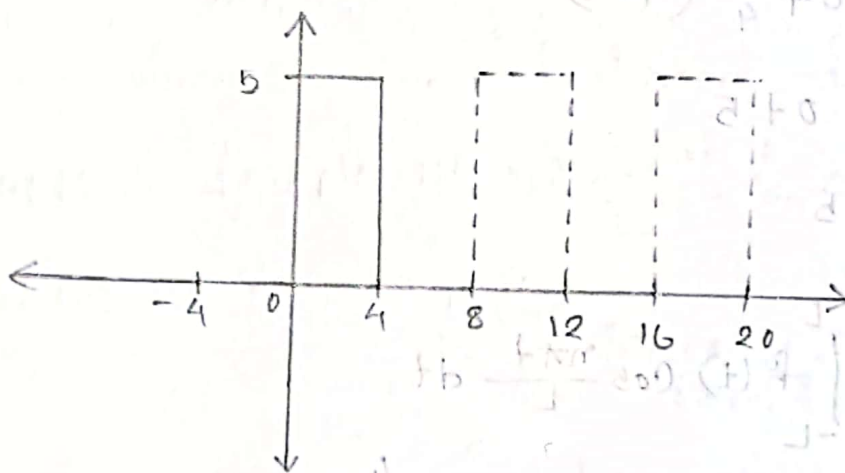
Fig: 08

Example - 22:  $y = f(t) = \begin{cases} 0 & ; -4 \leq t \leq 0 \\ 5 & ; 0 \leq t \leq 4 \end{cases}$

$f(t) = f(t+8)$  Hence  $T = 2L = 8$   $\therefore L = 4$

- a) Sketch the function for 3 cycles:  
 b) Find the Fourier series for the function.

a) Soln:



b) Soln: We know the Fourier series in the interval  $(-L, L)$  for  $f(t)$  is given by

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L} \right) \quad \text{--- (1)}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{4} \int_{-4}^4 f(t) dt$$

$$= \frac{1}{4} \int_{-4}^0 f(t) dt + \frac{1}{4} \int_0^4 f(t) dt$$

$$= \frac{1}{4} \int_{-4}^0 0 dt + \frac{1}{4} \int_0^4 5 dt$$

$$= 0 + \frac{5}{4} (4 - 0)$$

$$= 0 + 5$$

$$= 5$$

Again,

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{4} \int_{-4}^0 0 \cdot \cos \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 5 \cos \frac{n\pi t}{4} dt$$

$$= \frac{1}{4} \cdot 0 + \frac{5}{4} \cdot \frac{4}{n\pi} \left[ \sin \frac{n\pi t}{4} \right]_0^4$$

$$= 0 + \frac{5}{n\pi} (\sin n\pi - \sin 0)$$

$$= 0$$

Again,

$$\begin{aligned}
 b_n &= \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt \\
 &= \frac{1}{4} \int_{-4}^0 0 \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 5 \sin \frac{n\pi t}{4} dt \\
 &= 0 + \frac{5}{4} \cdot \frac{4}{n\pi} \left[ -\cos \frac{n\pi t}{4} \right]_0^4 \\
 &= \frac{5}{n\pi} [-\cos n\pi + \cos 0] \\
 &= \frac{-5}{n\pi} [\cos n\pi - 1]
 \end{aligned}$$

The Fourier series for above function is

$$\begin{aligned}
 f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \\
 &= \frac{5}{2} + \sum_{n=1}^{\infty} \left( 0 + \frac{-5}{n\pi} [\cos n\pi - 1] \sin \frac{n\pi x}{4} \right) \\
 &= \frac{5}{2} - \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} ([\cos n\pi - 1] \sin \frac{n\pi x}{4})
 \end{aligned}$$

(Am)

Problem: 10 (Only - Formula)

$$a \cos \theta + b \sin \theta = R \sin(\theta + \alpha)$$

$a \rightarrow$  Amplitude of the cosine wave

$b \rightarrow$  Amplitude of the sine wave

$R \rightarrow$  Amplitude of the new signal

$\theta \rightarrow$  Phase Shift.

$$\sin(A+B) = \sin A \cos B + \cos A \sin B \quad \text{--- (i)}$$

$$R \sin(\theta + \alpha) = R \sin \theta \cos \alpha + R \cos \theta \sin \alpha$$

$$a \cos \theta + b \sin \theta = R \cos \theta \sin \alpha + R \sin \theta \cos \alpha$$

$$a = R \sin \alpha \quad \text{--- (i)}$$

$$b = R \cos \alpha \quad \text{--- (ii)}$$

From (i) & (ii)

$$\frac{a}{b} = \frac{R \sin \alpha}{R \cos \alpha}$$

$$\text{or, } \frac{a}{b} = \tan \alpha$$

$$\text{or, } \tan \alpha = \frac{a}{b}$$

$$\text{or, } \alpha = \tan^{-1}(a/b)$$

Again,

from (i) & (ii) Squaring and then adding,

$$a^2 + b^2 = R^2 \sin^2 \alpha + R^2 \cos^2 \alpha$$

$$\text{or, } a^2 + b^2 = R^2 (\sin^2 \alpha + \cos^2 \alpha)$$

$$\text{or, } a^2 + b^2 = R^2 \cdot 1 \quad [\sin^2 \alpha + \cos^2 \alpha = 1]$$

$$\text{or, } a^2 + b^2 = R^2$$

$$\text{or, } R^2 = a^2 + b^2$$

$$\therefore R = \sqrt{a^2 + b^2}$$

### Example - 32

•  $x = \frac{1}{4} \cos 8t + \frac{1}{8} \sin 8t$ . What is the amplitude of the resultant signal?

Sol<sup>n</sup>: Given that,

$$x = \frac{1}{4} \cos 8t + \frac{1}{8} \sin 8t \quad \text{--- (i)}$$

Now comparing (i) with

$$a \cos \theta + b \sin \theta = R \sin(\theta + \alpha)$$

We get,

$$a = \frac{1}{4} \quad \& \quad b = \frac{1}{8}$$

Now,

the Required new signal,

$$R = \sqrt{a^2 + b^2}$$

$$\text{or, } R = \sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{1}{8}\right)^2}$$

$$\text{or, } R = \sqrt{\frac{1}{16} + \frac{1}{64}}$$

$$\text{or, } R = \sqrt{\frac{4+1}{64}}$$

$$\text{or, } R = \sqrt{\frac{5}{64}}$$

$$\therefore R = \frac{\sqrt{5}}{8}$$

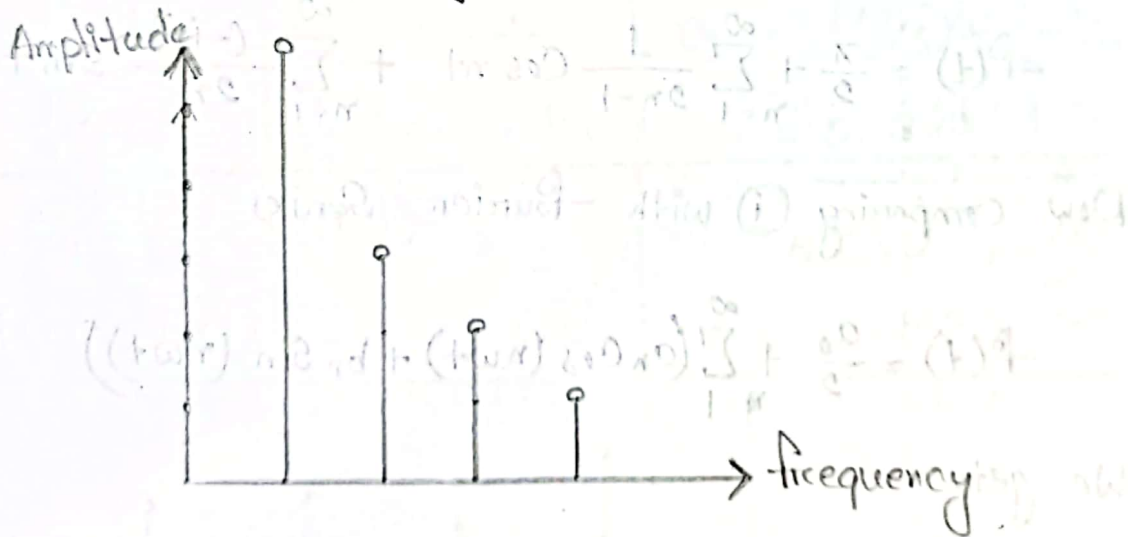
(Am)

### Problem - 12

Q: Define Line Spectrum or amplitude spectrum and phase spectrum (must answer).

Line Spectrum or amplitude spectrum: A plot of amplitude against angular frequency is called the line spectrum or amplitude spectrum.

Phase Spectrum: While that of phase against angular frequency is called the phase spectrum.



Ex-33 আমল না কিন্তু explanation জ্ঞান জানা দরকার

### Example - 34

• Plot the line spectrum (discrete frequency spectra) for the Fourier Series:

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos nt + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin t$$

Sol<sup>n</sup>: Given that,

The Fourier Series

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos nt + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin t \quad \text{--- (i)}$$

Now comparing (i) with Fourier Series

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t))$$

We get

$$a_n = \frac{1}{2n-1}$$

$$\text{and } b_n = \frac{(-1)^n}{2n}$$



We know that,

$$R = \sqrt{a^2 + b^2}$$

$$\therefore R_n = C_n = \sqrt{a_n^2 + b_n^2}$$

| $a_n = \frac{1}{2n-1}$ | $b_n = \frac{(-1)^n}{2n}$ | $C_n = \sqrt{a_n^2 + b_n^2}$                                   |
|------------------------|---------------------------|----------------------------------------------------------------|
| $a_1 = 1$              | $b_1 = \frac{1}{2}$       | $C_1 = \sqrt{(1)^2 + (1/2)^2}$<br>$= 1.118$                    |
| $a_2 = \frac{1}{3}$    | $b_2 = \frac{1}{4}$       | $C_2 = \sqrt{(\frac{1}{3})^2 + (\frac{1}{4})^2}$<br>$= 0.4167$ |
| $a_3 = \frac{1}{5}$    | $b_3 = -\frac{1}{6}$      | $C_3 = \sqrt{(\frac{1}{5})^2 + (-\frac{1}{6})^2}$<br>$= 0.260$ |
| $a_4 = \frac{1}{7}$    | $b_4 = \frac{1}{8}$       | $C_4 = \sqrt{(\frac{1}{7})^2 + (\frac{1}{8})^2}$<br>$= 0.190$  |

Again for frequency,

from eq<sup>n</sup> (i) we have the frequency =  $n\omega$

When  $n=1$  then the 1<sup>st</sup> frequency  $\omega = 1$

"  $n=2$  " " 2<sup>nd</sup> "  $2\omega = 2$

"  $n=3$  " " 3<sup>rd</sup> "  $3\omega = 3$

"  $n=4$  " " 4<sup>th</sup> "  $4\omega = 4$

Hence the required line spectrum is

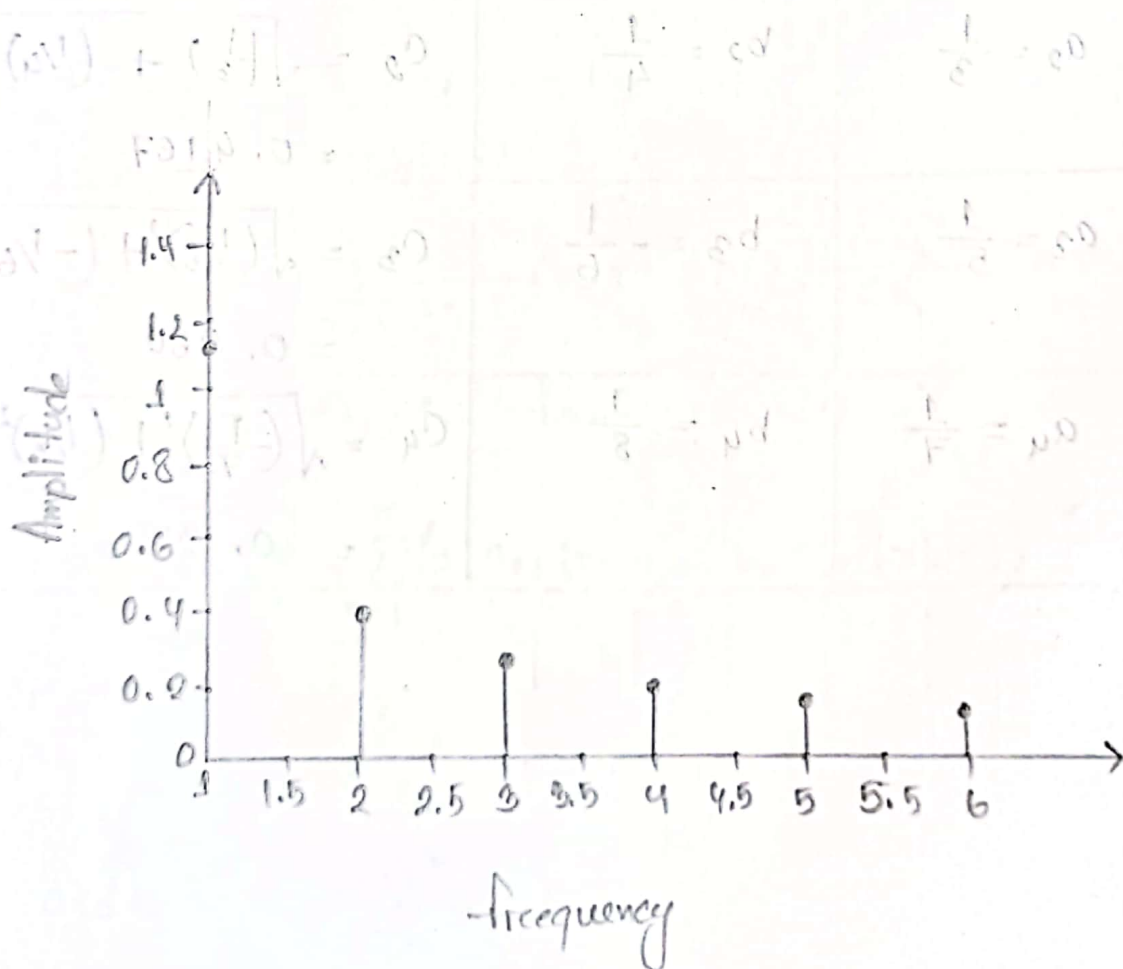


Fig: Line Spectrum

### Example - 35 (Home <sup>work</sup> ~~task~~)

- Line Spectrum : Example

$$S(t) = \frac{1}{2} + \frac{2}{\pi} \left( \cos(2\pi f_0 t) - \frac{1}{3} \cos(6\pi f_0 t) + \frac{1}{5} \cos(10\pi f_0 t) - \dots \right)$$

• Write the complex plot of Fourier Series.

Answer:-

The complex form of the Fourier series expansion of a periodic function  $f(t)$  of period  $T$ , is:-

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$

Where,

$$c_0 = \frac{1}{T} \int_{-L}^L f(t) dt$$

$$c_n = \frac{1}{T} \int_{-L}^L f(t) e^{-in\omega t} dt$$

$$c_n^* = c_{-n} = \frac{1}{T} \int_{-L}^L f(t) e^{in\omega t} dt$$

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Example-4

$$f(t) = 3t; \quad -1 \leq t < 1$$

$f(t) = f(t+2)$ ; [This indicates it is periodic with period 2]

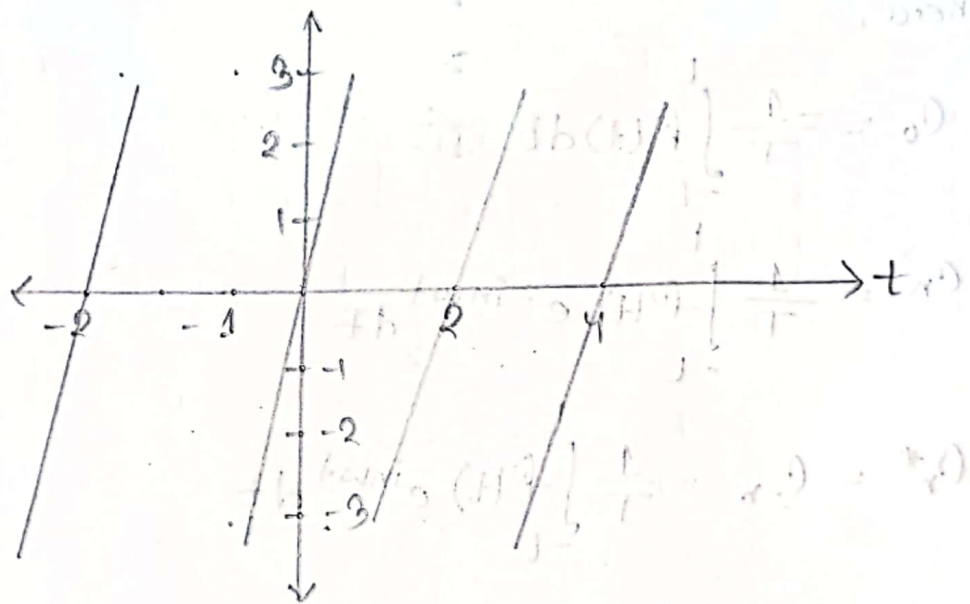
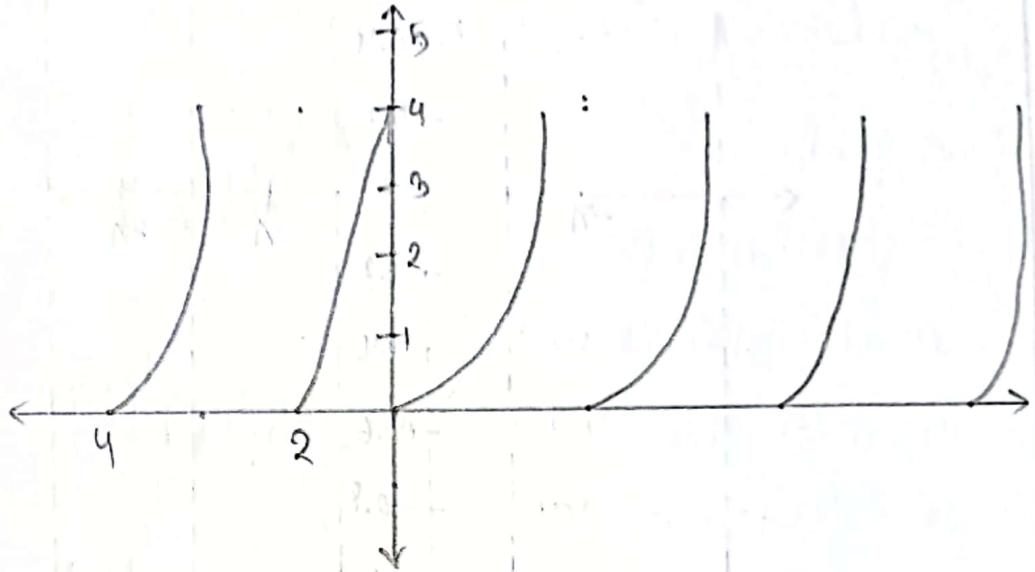


Fig: 06,  $f(t) = 3t$ , Period  $T = 2$

### Example-05

$$f(t) = t^2 \quad ; \quad 0 \leq t < 2$$

$$f(t) = f(t+2) \quad [\text{Indicating it is periodic with period 2}]$$



### Example - 07

$$f(t) = \begin{cases} -1 & \text{if } 0 \leq t < \frac{\pi}{2} \\ 1 & \text{if } \frac{\pi}{2} \leq t < \frac{3\pi}{2} \\ -1 & \text{if } \frac{3\pi}{2} \leq t < 2\pi \end{cases}$$

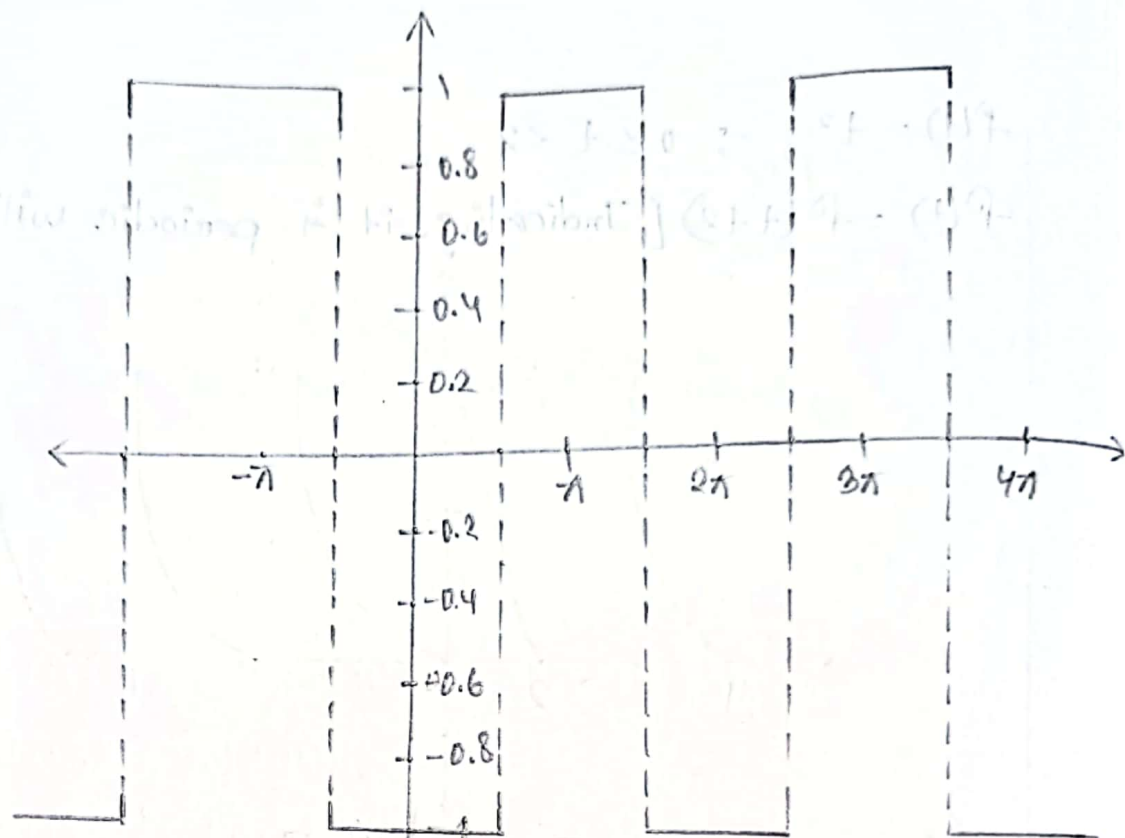


Fig: 09, (Period  $T = 2\pi$ ).

### Example -10

$$f(t) = A \sin(\omega t + \phi)$$

If  $A=1$ ;  $\phi=0$ , then draw the graph of  $f(t)=?$

Soln:

Given that,

$$f(t) = A \sin(\omega t + \phi)$$

$$\text{or, } f(t) = 1 \cdot \sin(\omega t + 0)$$

$$\therefore f(t) = \sin \omega t$$

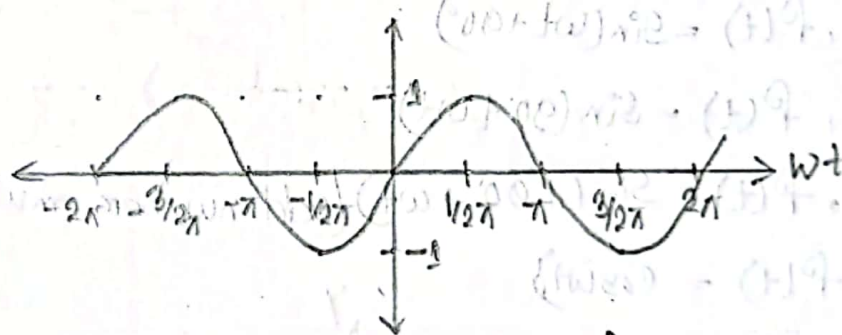


Fig: 17 ( $f(t) = A \sin \omega t$ )

### Example - 11

$$f(t) = A \sin(\omega t + \phi)$$

If  $A = 1$ ;  $\phi = 90^\circ$ , then draw the graph of  $f(t) = ?$

Soln:

Given that,

$$f(t) = A \sin(\omega t + \phi)$$

$$\text{or, } f(t) = 1 \sin(\omega t + \phi)$$

$$\text{or, } f(t) = 1 \sin(\omega t + 90^\circ)$$

$$\text{or, } f(t) = \sin(\omega t + 90^\circ)$$

$$\text{or, } f(t) = \sin(90^\circ + \omega t)$$

$$\text{or, } f(t) = \sin(1 \cdot 90^\circ + \omega t) \text{ [odd number multiplication with } 90^\circ]$$

$$\therefore f(t) = \cos(\omega t)$$

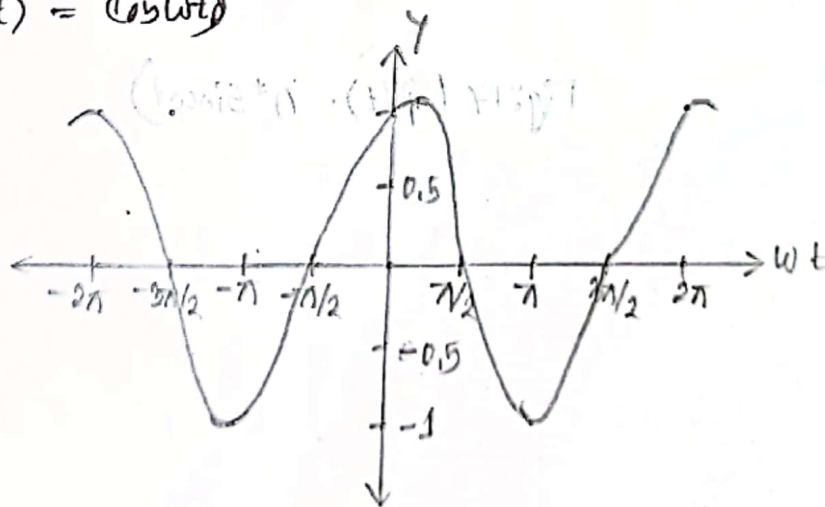


Fig-18 ( $f(t) = A \cos \omega t$ )

### Example-12

$$f(t) = A \sin(\omega t + \phi)$$

$$f(t) = a \cos \omega t + b \sin \omega t$$

### Example-19

• Determine the value of Fourier coefficient  $a_0$

Sol<sup>n</sup>: We know that,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t)) \quad \text{--- (i)}$$

Integrate (i) on both sides from  $-\pi$  to  $\pi$  i.e. period  $= T = 2\pi$ .

We have

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

$$\begin{aligned} \int_{-\pi}^{\pi} f(t) dt &= \int_{-\pi}^{\pi} \left[ \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t)) \right] dt \\ &= \frac{a_0}{2} \int_{-\pi}^{\pi} dt + \sum_{n=1}^{\infty} \left[ a_n \int_{-\pi}^{\pi} \cos(n\omega t) dt + b_n \int_{-\pi}^{\pi} \sin(n\omega t) dt \right] \\ &= \frac{a_0}{2} [t]_{-\pi}^{\pi} + \sum_{n=1}^{\infty} \left[ a_n \left[ \frac{\sin(n\omega t)}{n\omega} \right]_{-\pi}^{\pi} - b_n \left[ \frac{\cos(n\omega t)}{n\omega} \right]_{-\pi}^{\pi} \right] \end{aligned}$$

$$\begin{aligned}
 &= \frac{a_0}{2} [\pi - (-\pi)] + \sum_{n=1}^{\infty} \left[ a_n \left[ \frac{\sin(n\omega t)}{n\omega} \right]_{-\pi}^{\pi} - b_n \left[ \frac{\cos(n\omega t)}{n\omega} \right]_{-\pi}^{\pi} \right] \\
 &= \frac{a_0}{2} [\pi + \pi] + \sum_{n=1}^{\infty} \left[ a_n \left[ \frac{\sin(n\omega t)}{n\omega} \right]_{-\pi}^{\pi} - b_n \left[ \frac{\cos(n\omega t)}{n\omega} \right]_{-\pi}^{\pi} \right] \\
 &= \frac{a_0}{2} 2\pi + \sum_{n=1}^{\infty} \left[ \frac{a_n}{n\omega} (\sin(n\omega\pi) - \sin(-n\omega\pi)) - \frac{b_n}{n\omega} (\cos(n\omega\pi) - \cos(-n\omega\pi)) \right]
 \end{aligned}$$

$$= \pi a_0 + \sum_{n=1}^{\infty} \left[ \frac{a_n}{n\omega} (\sin(n\omega\pi) + \sin(n\omega\pi)) - \frac{b_n}{n\omega} (\cos(n\omega\pi) - \cos(n\omega\pi)) \right]$$

$$= \pi a_0 + \sum_{n=1}^{\infty} \left[ \frac{a_n}{n\omega} 2\sin(n\omega\pi) \right] = 0 \quad \text{--- (ii)}$$

When  $n=1$  &  $\omega=1$

Then,

$$\frac{a_n}{n\omega} 2\sin(n\omega\pi) = \frac{a_1}{1} 2\sin(\pi) = 0$$

$n=2$

$$\frac{a_n}{n\omega} 2\sin(n\omega\pi) = \frac{a_2}{2} 2\sin(2\pi) = 0$$

Similarly,

$$\frac{a_n}{n\omega} 2\sin(n\omega\pi) = 0 \quad \text{for } n=1,2,3$$

from (ii)

$$\therefore \int_{-\pi}^{\pi} f(t) dt = \pi a_0 + \sum_{n=1}^{\infty} \left[ \frac{a_n}{n\omega} 2\sin(n\omega\pi) \right] = 0$$

$$\text{or, } \int_{-\pi}^{\pi} f(t) dt = \pi a_0 + 0$$

$$\text{or, } \int_{-\pi}^{\pi} f(t) dt = \pi a_0$$

$$\therefore a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt.$$

### Example - 23

$$f(t) = -1 \quad ; \quad -\pi < t < 0$$

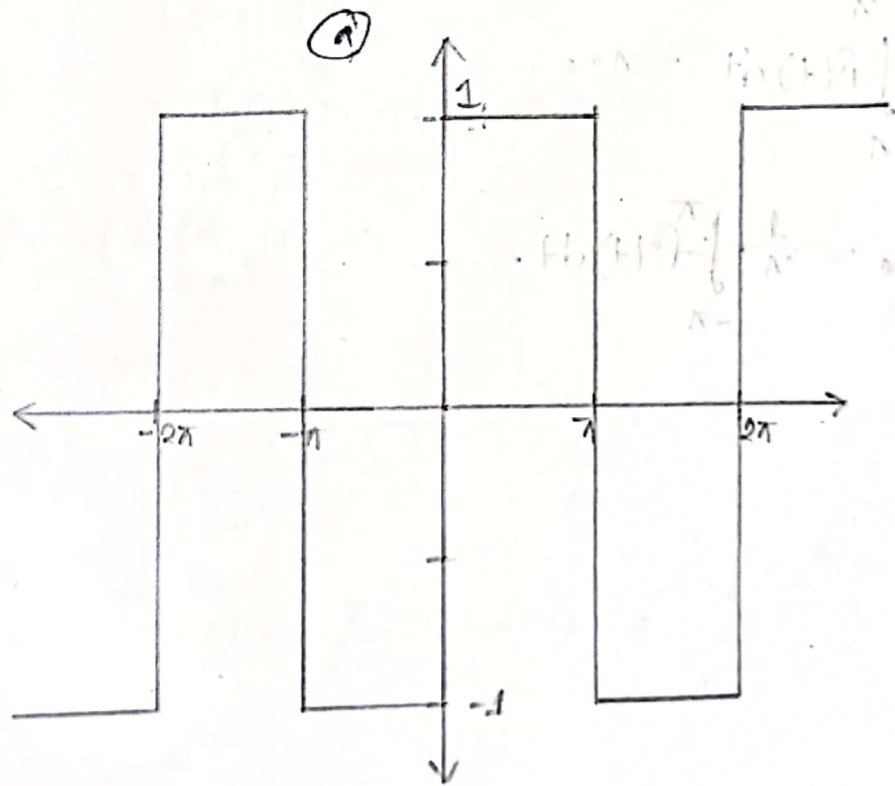
$$= 1 \quad ; \quad 0 < t < \pi$$

$$f(t) = f(t + 2\pi) \quad \text{Here } T = 2L = 2\pi \quad \therefore L = \pi$$

(a) Sketch the function for 3 cycles.

(b) Find the Fourier series for the function.

Sol/no



(b)

We know the Fourier series in the interval  $(-L, L)$   
for  $f(t)$  is given by -

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L} \right) \quad \text{--- (i)}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(t) dt + \frac{1}{\pi} \int_0^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-1) dt + \frac{1}{\pi} \int_0^{\pi} (1) dt$$

$$= -\frac{1}{\pi} [t]_{-\pi}^0 + \frac{1}{\pi} [t]_0^{\pi}$$

$$= -\frac{1}{\pi} [0 - (-\pi)] + \frac{1}{\pi} [\pi - 0]$$

$$= -\frac{1}{\pi} [\pi] + \frac{1}{\pi} [\pi]$$

$$= -1 + 1$$

$$= 0$$

$$a_n = \frac{1}{L} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-1) \cos nt dt + \frac{1}{\pi} \int_0^{\pi} (1) \cos nt dt$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 \cos nt dt + \frac{1}{\pi} \int_0^{\pi} \cos nt dt$$

$$= -\frac{1}{\pi} \times \frac{1}{n} [\sin nt]_{-\pi}^0 + \frac{1}{\pi} \times \frac{1}{n} [\sin nt]_0^{\pi}$$

$$= -\frac{1}{\pi n} [\sin 0 - \sin(-n\pi)] + \frac{1}{\pi n} [\sin \pi - 0]$$

$$= -\frac{1}{\pi n} [0 - \sin(-n\pi)] + \frac{1}{\pi n} [\sin \pi - 0]$$

$$= -\frac{1}{\pi n} [0 + \sin(n\pi)] + \frac{1}{\pi n} [\sin n\pi - 0]$$

$$= -\frac{1}{\pi n} [0 + 0] + \frac{1}{\pi n} [0 - 0]$$

$$= 0$$

Again,

$$\begin{aligned}b_n &= \frac{1}{L} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{L} dt \\&= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt \\&= \frac{1}{\pi} \int_{-\pi}^0 f(t) \sin \frac{n\pi t}{\pi} dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt \\&= \frac{1}{\pi} \int_{-\pi}^0 f(t) \sin nt dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin nt dt \\&= \frac{1}{\pi} \int_{-\pi}^0 (-1) \sin nt dt + \frac{1}{\pi} \int_0^{\pi} (1) \sin nt dt \\&= -\frac{1}{\pi} \int_{-\pi}^0 \sin nt dt + \frac{1}{\pi} \int_0^{\pi} \sin nt dt \\&= -\frac{1}{\pi} \times \frac{1}{n} [-\cos nt]_{-\pi}^0 + \frac{1}{\pi} \times \frac{1}{n} [-\cos nt]_0^{\pi} \\&= \frac{1}{\pi n} [\cos nt]_{-\pi}^0 + \frac{1}{\pi n} [-\cos nt]_0^{\pi} \\&= \frac{1}{\pi n} [\cos 0 - \cos(-n\pi)] - \frac{1}{\pi n} [\cos n\pi - \cos 0] \\&= \frac{1}{\pi n} [1 - \cos n\pi] - \frac{1}{\pi n} [\cos n\pi - 1] \\&= \frac{1}{\pi n} - \frac{1}{\pi n} \cos n\pi - \frac{1}{\pi n} \cos n\pi + \frac{1}{\pi n} \\&= \frac{2}{\pi n} - \frac{2}{\pi n} \cos n\pi \\&= \frac{2}{\pi n} (1 - \cos n\pi)\end{aligned}$$



The Fourier series for the above function is

$$\begin{aligned}
 f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L} \\
 &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{\pi} \\
 &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt \\
 &= 0 + \sum_{n=1}^{\infty} 0 \cdot \cos nt + \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) \sin nt \\
 &= 0 + \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) \sin nt
 \end{aligned}$$

(Am)

$$\int_0^{\pi} [f(x) \cos nx] \frac{1}{\pi} dx = \int_0^{\pi} [f(x) \sin nx] \frac{1}{\pi} dx$$

$$\int_0^{\pi} [f(x) \cos nx] \frac{1}{\pi} dx = \int_0^{\pi} [f(x) \sin nx] \frac{1}{\pi} dx$$

$$[0 \cos nx - \pi \sin nx] \frac{1}{\pi} = [\pi \cos nx - 0 \sin nx] \frac{1}{\pi}$$

$$[1 - \pi \cos nx] \frac{1}{\pi} = [\pi \cos nx - 1] \frac{1}{\pi}$$

$$\frac{1}{\pi} - \pi \cos nx \frac{1}{\pi} = \pi \cos nx \frac{1}{\pi} - \frac{1}{\pi}$$

$$\pi \cos nx \frac{1}{\pi} = \frac{1}{\pi}$$

$$(\pi \cos nx) \frac{1}{\pi} = \frac{1}{\pi}$$



### Example-24

$$y = f(t) = -t ; -\pi \leq t < 0$$
$$= t ; 0 \leq t < \pi$$

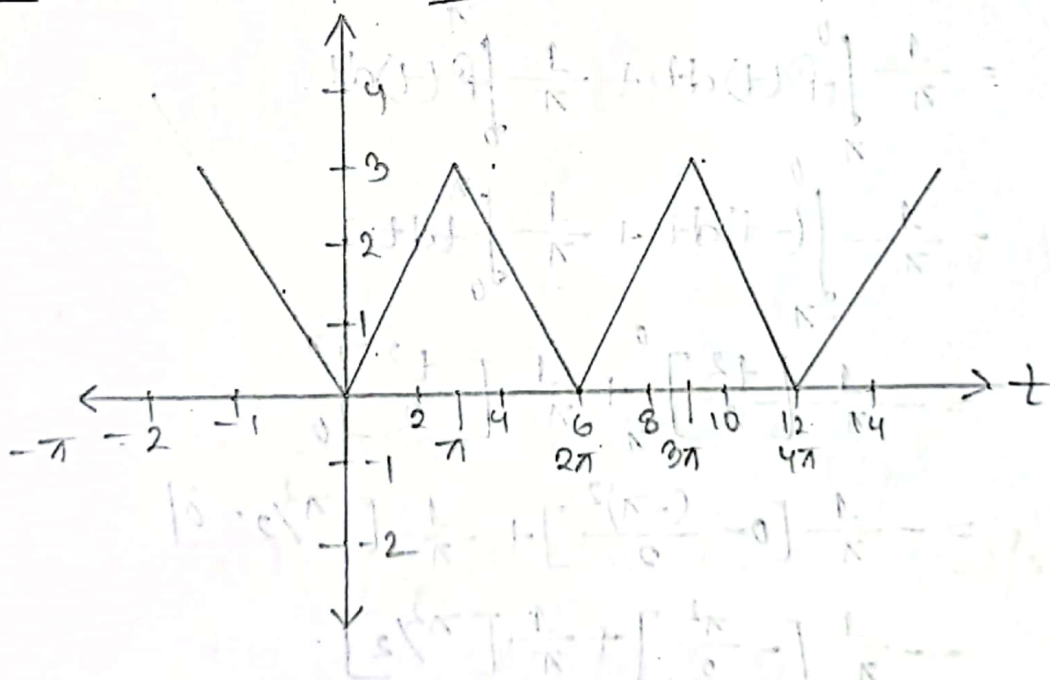
$$f(t) = f(t + 2\pi) \text{ Hence, } T = 2L = 2\pi \therefore L = \pi$$

① sketch the function for 3 cycles.

② Find the Fourier Series for the function.

Soln:

①



(b)

Sol<sup>n</sup>: We know the Fourier series in the interval  $(-L, L)$  for  $f(t)$  is given by,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L} \right) \quad \text{--- (i)}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 f(t) dt + \frac{1}{\pi} \int_0^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-t) dt + \frac{1}{\pi} \int_0^{\pi} t dt$$

$$= -\frac{1}{\pi} \left[ \frac{t^2}{2} \right]_{-\pi}^0 + \frac{1}{\pi} \left[ \frac{t^2}{2} \right]_0^{\pi}$$

$$= -\frac{1}{\pi} \left[ 0 - \frac{(-\pi)^2}{2} \right] + \frac{1}{\pi} \left[ \frac{\pi^2}{2} - 0 \right]$$

$$= -\frac{1}{\pi} \left[ -\frac{\pi^2}{2} \right] + \frac{1}{\pi} \left[ \frac{\pi^2}{2} \right]$$

$$= \frac{1}{\pi} \times \frac{\pi^2}{2} + \frac{1}{\pi} \times \frac{\pi^2}{2}$$

$$= \frac{\pi}{2} + \frac{\pi}{2}$$

$$= \frac{2\pi}{2} = \pi$$



Again,

$$\begin{aligned}
 a_n &= \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt \\
 &= \frac{1}{\pi} \int_{-\pi}^0 f(t) \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} f(t) \cos(nt) dt \\
 &= \frac{1}{\pi} \int_{-\pi}^0 (-t) \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt \\
 &= -\frac{1}{\pi} \int_{-\pi}^0 t \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt \quad \text{--- (ii)}
 \end{aligned}$$

Now,

$$I = \int t \cos(nt) dt$$

$$= t \int \cos(nt) dt - \int \left[ \frac{d}{dt} (t) \int \cos(nt) dt \right] dt$$

$$\left[ \because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx \right]$$

$$= t \frac{\sin(nt)}{n} - \int 1 \cdot \frac{\sin(nt)}{n} dt$$

$$= \frac{t}{n} \sin(nt) - \frac{1}{n} \int \sin(nt) dt$$

$$= \frac{t}{n} \sin(nt) - \frac{1}{n} \frac{(-\cos(nt))}{n}$$

$$= \frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \text{ --- (iii)}$$

Putting the value (iii) & (ii)

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 t \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt$$

$$= -\frac{1}{\pi} \left[ \frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_{-\pi}^0 + \frac{1}{\pi} \left[ \frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_0^{\pi}$$

$$= -\frac{1}{\pi} \left[ \frac{0}{n} \sin(0) + \frac{1}{n^2} \cos(0) - \frac{(-\pi)}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] + \frac{1}{\pi} \left[ \frac{\pi}{n} \sin(n\pi) + \frac{1}{n^2} \cos(n\pi) - \frac{0}{n} \sin(0) - \frac{1}{n^2} \cos(0) \right]$$

$$= -\frac{1}{\pi} \left[ 0 + \frac{1}{n^2} \times 1 + \frac{\pi}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] + \frac{1}{\pi} \left[ \frac{\pi}{n} \times 0 + \frac{1}{n^2} \cos(n\pi) - 0 - \frac{1}{n^2} \times 1 \right]$$

$$= -\frac{1}{\pi} \left[ \frac{1}{n^2} - \frac{\pi}{n} \sin(n\pi) - \frac{1}{n^2} \cos(n\pi) \right] + \frac{1}{\pi} \left[ \frac{1}{n^2} \cos(n\pi) - \frac{1}{n^2} \right]$$

$$= -\frac{1}{\pi} \left[ \frac{1}{n^2} - \frac{\pi}{n} \times 0 - \frac{1}{n^2} \cos(n\pi) \right] + \frac{1}{\pi} \left[ \frac{1}{n^2} \cos(n\pi) - \frac{1}{n^2} \right]$$

$$= -\frac{1}{\pi n^2} + \frac{1}{\pi n^2} \cos(n\pi) + \frac{1}{\pi n^2} \cos(n\pi) - \frac{1}{\pi n^2}$$

$$= \frac{2}{\pi n^2} \cos n\pi - \frac{2}{\pi n^2}$$

$$= \frac{2}{\pi n^2} (\cos n\pi - 1)$$

Again,

$$b_n = \frac{1}{L} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt$$

$$= \frac{1}{\pi} \left( \int_{-\pi}^0 f(t) \sin(nt) dt + \int_0^{\pi} f(t) \sin(nt) dt \right)$$

$$= \frac{1}{\pi} \int_{-\pi}^0 (-t) \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 t \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt \quad \text{--- (iv)}$$

Now,

$$\int t \sin(nt) dt = t \int \sin(nt) dt - \int \left\{ \frac{d}{dt} (t) \int \sin(nt) dt \right\} dt$$

$$\left[ \because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx \right]$$

$$= t \left( \frac{-\cos(nt)}{n} \right) - \int 1 \cdot \frac{-\cos(nt)}{n} dt$$

$$= -\frac{t}{n} \cos(nt) + \frac{1}{n} \int \cos(nt) dt$$

$$= -\frac{t}{n} \cos(nt) + \frac{1}{n} \cdot \frac{\sin(nt)}{n}$$

$$= -\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \quad \text{--- (v)}$$

Putting the value of (v) in (iv)

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt$$

$$= -\frac{1}{\pi} \int_{-\pi}^0 t \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt$$

$$= -\frac{1}{\pi} \left[ \frac{-t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_{-\pi}^0 + \frac{1}{\pi} \left[ \frac{-t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_0^{\pi}$$

$$= -\frac{1}{\pi} \left[ \frac{-0}{n} \cos(0) + \frac{1}{n^2} \sin(0) - \frac{-(+\pi)}{n} \cos(-n\pi) - \frac{1}{n^2} \sin(-n\pi) \right] + \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos(n\pi) + \right.$$

$$\begin{aligned}
& -\frac{1}{n^2} \sin(n\pi) - \frac{-(0)}{n} \cos(0) - \frac{1}{n^2} \sin(0) \\
& = -\frac{1}{\pi} \left[ 0 + 0 - \frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) \right] + \\
& \quad \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) + 0 - 0 \right] \\
& = -\frac{1}{\pi} \left[ -\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \times 0 \right] + \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \times 0 \right] \\
& = \frac{\pi}{\pi n} \cos(n\pi) - \frac{\pi}{\pi n} \cos(n\pi) \\
& = 0
\end{aligned}$$

The Fourier series for the above function is:

$$\begin{aligned}
f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}\right)t + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}\right)t \\
&= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} (\cos n\pi - 1) \cos\left(\frac{n\pi}{L}\right)t + 0 \\
&= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} (\cos n\pi - 1) \cos\left(\frac{n\pi}{\pi}\right)t \\
&= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi} (\cos n\pi - 1) \cos nt \\
& \quad \quad \quad \underline{\underline{(A_m)}}
\end{aligned}$$

Example-25

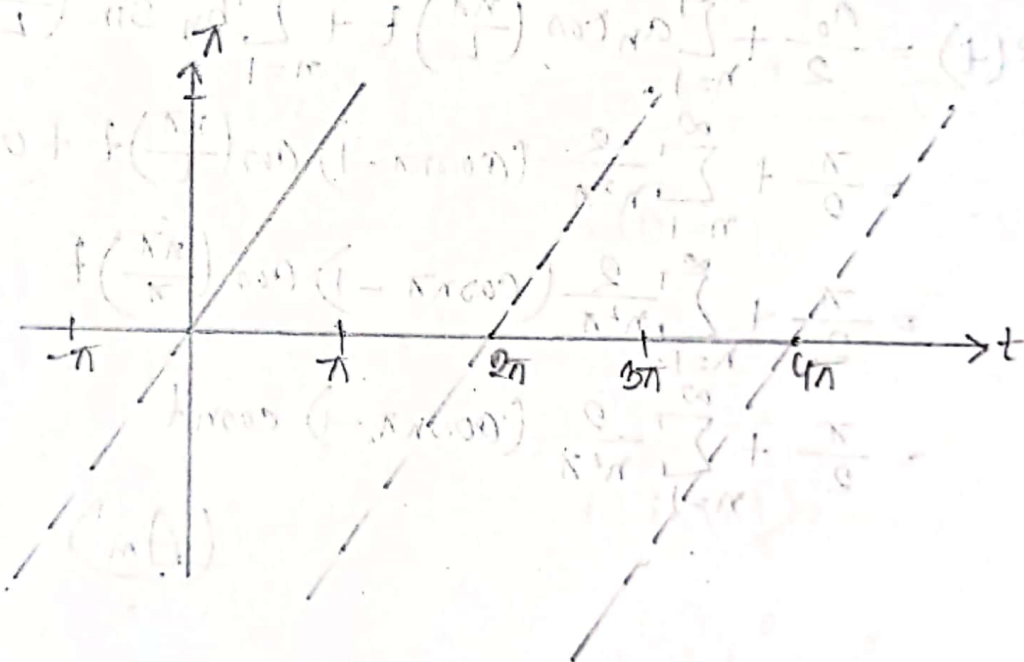
$$y = f(t) = t; -\pi \leq t \leq \pi$$

Here  $T = 2L = 2\pi \therefore L = \pi$

- (a) Sketch the function for 3 cycles  
(b) Find the Fourier series for the function.

(a)

Soln:



(b)

Sol<sup>n</sup>: We know the Fourier series in the interval  $(-L, L)$  for  $f(t)$  is given by,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L} \right) \quad \text{--- (1)}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} t dt \quad [\text{from eqn (1)}]$$

$$= \frac{1}{\pi} \left[ \frac{t^2}{2} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[ \frac{\pi^2}{2} - \frac{(-\pi)^2}{2} \right]$$

$$= \frac{1}{\pi} \left[ \frac{\pi^2}{2} - \frac{\pi^2}{2} \right]$$

$$= 0$$

again,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t \cos nt dt \quad [\text{From eqn (i)}]$$

Now

$$\int t \cos nt dt = t \int \cos nt dt - \int \left\{ \frac{d}{dt} (t) \cos nt \right\} dt$$

$$= \frac{t}{n} \times \sin nt - \int 1 \cdot \frac{\sin nt}{n} dt$$

$$= \frac{t}{n} \sin nt - \frac{1}{n} \cdot \frac{1}{n} (-\cos nt)$$

$$= \frac{t}{n} \sin nt + \frac{1}{n^2} \cos (nt) \quad \text{--- (ii)}$$

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} t \cos nt dt$$

$$= \frac{1}{\pi} \left[ \frac{t}{n} \sin nt + \frac{1}{n^2} \cos nt \right]_{-\pi}^{\pi} \quad [\text{From (ii)}]$$

$$= \frac{1}{\pi} \left[ \frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \left\{ \frac{-\pi}{n} \sin(-n\pi) + \frac{1}{n^2} \cos(-n\pi) \right\} \right]$$

$$= \frac{1}{\pi} \left[ \frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi + \frac{\pi}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right]$$

$$= \frac{1}{\pi} \left[ \frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi + \frac{\pi}{n} (-\sin n\pi) - \frac{1}{n^2} \cos(n\pi) \right]$$

$$\left[ \because \sin(-\theta) = -\sin\theta \right. \\ \left. \cos(-\theta) = \cos\theta \right]$$

$$= \frac{1}{\pi} \left[ \cancel{\frac{\pi}{n} \sin n\pi} + \cancel{\frac{1}{n^2} \cos n\pi} - \cancel{\frac{\pi}{n} \sin n\pi} - \cancel{\frac{1}{n^2} \cos n\pi} \right]$$

$$= 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t \sin nt dt \quad (0 \rightarrow \pi, -\pi \rightarrow 0)$$

Now,

$$\int t \sin nt dt = t \int \sin nt dt - \int \left\{ \frac{d}{dt}(t) \int \sin nt dt \right\} dt$$

$$= t \cdot \frac{1}{n} (-\cos nt) - \int 1 \cdot \frac{1}{n} (-\cos nt) dt$$

$$= -\frac{t}{n} \cos nt + \frac{1}{n} \int \cos nt dt$$

$$= -\frac{t}{n} \cos nt + \frac{1}{n^2} \sin nt \quad \text{--- (ii)}$$

$$\begin{aligned}
 \therefore b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} t \cdot \sin nt \cdot dt \\
 &= \frac{1}{\pi} \left[ -\frac{t}{n} \cos nt + \frac{1}{n^2} \sin nt \right]_{-\pi}^{\pi} \quad [\text{from (iii)}] \\
 &= \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \left\{ -\frac{-\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin(-n\pi) \right\} \right] \\
 &= \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \left\{ \frac{\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin(-n\pi) \right\} \right] \\
 &= \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos(-n\pi) - \frac{1}{n^2} \sin(-n\pi) \right]
 \end{aligned}$$

$$\begin{aligned}
 [\because \sin(-\theta) &= -\sin \theta, \\
 \cos(-\theta) &= \cos \theta]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi \right] \\
 &= \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi - \frac{\pi}{n} \cos n\pi \right] \\
 &= -\frac{1}{\pi} \times \frac{2\pi}{n} \cos n\pi \\
 &= -\frac{2}{n} \cos n\pi
 \end{aligned}$$

The Fourier series for the above function is:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi t + \sum_{n=1}^{\infty} b_n \sin n\pi t$$

$$= 0 + 0 + \sum_{n=1}^{\infty} b_n \sin n\pi t$$

$$= \sum_{n=1}^{\infty} -\frac{2}{n} \cos n\pi \cdot \sin n\pi t$$

$$= 0 + -2 \sum_{n=1}^{\infty} \frac{1}{n} \cos n\pi \cdot \sin n\pi t.$$

Example-26

$$y = f(t) = 0 \quad ; \quad -1 \leq t \leq -\frac{1}{2}$$

$$= \cos 3\pi t \quad ; \quad -\frac{1}{2} \leq t \leq \frac{1}{2}$$

$$= 0 \quad ; \quad \frac{1}{2} \leq t \leq 1$$

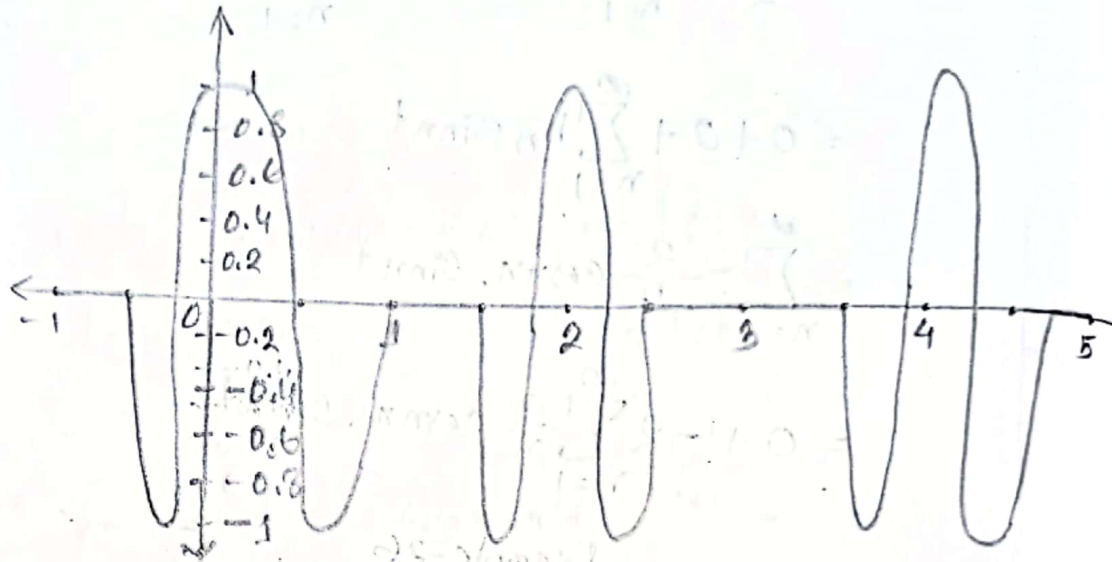
$$f(t) = f(t+1) \quad \text{Here } T = 2L = 2 \therefore L = 1$$

(A) Sketch the function for 3 cycles

(B) Find the Fourier series for the function.

(a)

Sol<sup>n</sup>:-



(b)

Sol<sup>n</sup>:- We know the Fourier series in the interval  $(-L, L)$  for  $f(t)$  is given by,

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{2} \int_{-1}^1 f(t) dt$$

$$= \int_{-1}^{-1/2} f(t) dt + \int_{-1/2}^{1/2} f(t) dt + \int_{1/2}^1 f(t) dt$$

$$= 0 + \int_{-1/2}^{1/2} \cos 3\pi t dt + 0$$

$$= \int_{-1/2}^{1/2} \cos 3\pi t \, dt$$

$$= \frac{1}{3\pi} [\sin 3\pi t]_{-1/2}^{1/2}$$

$$= \frac{1}{3\pi} \left[ \sin 3\pi \times \frac{1}{2} - \sin \left( -\frac{3\pi}{2} \right) \right]$$

$$= \frac{1}{3\pi} \left[ \sin \frac{3\pi}{2} + \sin \frac{3\pi}{2} \right]$$

$$= \frac{1}{3\pi} \times 2 \sin \frac{3\pi}{2}$$

$$= \frac{2}{3\pi} \sin \frac{3\pi}{2}$$

$$= \frac{2}{3\pi} (-1)$$

$$= \frac{-2}{3\pi}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} \, dt$$

$$= \int_{-1}^1 f(t) \cos \frac{n\pi t}{1} \, dt$$

$$= \int_{-1}^1 f(t) \cos n\pi t \, dt$$



$$= \int_{-1}^{-1/2} f(t) \cos n\pi t dt + \int_{-1/2}^{1/2} f(t) \cos n\pi t dt + \int_{1/2}^1 f(t) \cos n\pi t dt$$

$$= \int_{-1}^{-1/2} 0 \cdot \cos n\pi t dt + \int_{-1/2}^{1/2} f(t) \cos n\pi t dt + \int_{1/2}^1 0 \cdot \cos n\pi t dt$$

$$= \int_{-1}^{-1/2} 0 \cdot \cos n\pi t dt + \int_{-1/2}^{1/2} f(t) \cos n\pi t dt + \int_{1/2}^1 0 \cdot \cos n\pi t dt$$

$$= \int_{-1/2}^{1/2} f(t) \cos n\pi t dt$$

$$= \int_{-1/2}^{1/2} \cos 3\pi t \cos n\pi t dt$$

$$= \frac{1}{2} \int_{-1/2}^{1/2} 2 \cos 3\pi t \cos n\pi t dt$$

$$= \frac{1}{2} \int_{-1/2}^{1/2} \{ \cos(3\pi t + n\pi t) + \cos(3\pi t - n\pi t) \} dt$$

$$= \frac{1}{2} \int_{-1/2}^{1/2} \cos(3\pi t + n\pi t) dt + \frac{1}{2} \int_{-1/2}^{1/2} \cos(3\pi t - n\pi t) dt$$

$$= -\frac{1}{2} \int_{-1/2}^{1/2} \cos(3\pi + n\pi) t dt + \frac{1}{2} \int_{-1/2}^{1/2} \cos(3\pi - n\pi) t dt$$

$$= \frac{1}{2} \frac{1}{(3\pi + n\pi)} \left[ \sin(3\pi + n\pi) t \right]_{-1/2}^{1/2} + \frac{1}{2} \frac{1}{(3\pi - n\pi)} \left[ \sin(3\pi - n\pi) t \right]_{-1/2}^{1/2}$$

$$= \frac{1}{2} \frac{1}{(3\pi + n\pi)} \left[ \sin(3+n)\pi t \right]_{-1/2}^{1/2} + \frac{1}{2} \frac{1}{(3\pi - n\pi)} \left[ \sin(3-n)\pi t \right]_{-1/2}^{1/2}$$

$$= \frac{1}{2} \frac{1}{(3\pi + n\pi)} \left[ \sin(3+n) \frac{\pi}{2} - \sin(3+n) \left(-\frac{\pi}{2}\right) \right] + \frac{1}{2(3\pi - n\pi)} \left[ \sin(3-n) \frac{\pi}{2} - \sin(3-n) \left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{1}{2} \frac{1}{(3\pi + n\pi)} \left[ \sin(3+n) \frac{\pi}{2} + \sin(3+n) \frac{\pi}{2} \right] + \frac{1}{2(3\pi - n\pi)} \left[ \sin(3-n) \frac{\pi}{2} + \sin(3-n) \frac{\pi}{2} \right]$$

$$= \frac{1}{2} \frac{2}{(3\pi + n\pi)} \sin(3+n) \frac{\pi}{2} + \frac{1}{2} \times \frac{2}{(3\pi - n\pi)} \sin(3-n) \frac{\pi}{2}$$

$$= \frac{1}{(3+n)\pi} \sin(3+n) \frac{\pi}{2} + \frac{1}{(3-n)\pi} \sin(3-n) \frac{\pi}{2}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{1} \int_{-1}^1 f(t) \sin \frac{n\pi t}{1} dt$$

$$= \int_{-1}^{-1/2} f(t) \sin n\pi t dt + \int_{-1/2}^{1/2} f(t) \sin n\pi t dt + \int_{1/2}^1 f(t) \sin n\pi t dt$$

$$= \int_{-1}^{-1/2} 0 \cdot \sin n\pi t dt + \int_{-1/2}^{1/2} f(t) \sin n\pi t dt + \int_{1/2}^1 0 \cdot \sin n\pi t dt$$

$$= \int_{-1/2}^{1/2} f(t) \sin n\pi t dt$$

$$= \int_{-1/2}^{1/2} \cos 3\pi t + \sin n\pi t dt \quad [\text{from (i)}]$$

$$= \frac{1}{2} \int_{-1/2}^{1/2} 2 \cos 3\pi t + \sin n\pi t dt$$

$$= \frac{1}{2} \int_{-1/2}^{1/2} \{ \sin(3\pi t + n\pi t) - \sin(3\pi t - n\pi t) \} dt \quad [\because 2 \cos A \sin B = \sin(A+B) - \sin(A-B)]$$

$$= \frac{1}{2} \int_{-1/2}^{1/2} \sin(3\pi + n\pi) t dt - \frac{1}{2} \int_{-1/2}^{1/2} \sin(3\pi - n\pi) t dt$$

$$= \frac{1}{2} \frac{1}{(3\pi + n\pi)} \left[ -\cos(3\pi + n\pi) t \right]_{-1/2}^{1/2} - \frac{1}{2} \frac{1}{(3\pi - n\pi)} \left[ -\cos(3\pi - n\pi) t \right]_{-1/2}^{1/2}$$

$$= \frac{1}{2(3\pi + n\pi)} \left[ -\cos(3\pi + n\pi) \frac{1}{2} + \cos(3\pi + n\pi) \left(-\frac{1}{2}\right) \right] -$$

$$\frac{1}{2(3\pi - n\pi)} \left[ -\cos(3\pi - n\pi) \frac{1}{2} + \cos(3\pi - n\pi) \left(-\frac{1}{2}\right) \right]$$

$$= \frac{1}{2(3\pi + n\pi)} \left[ -\cos(3+n) \frac{\pi}{2} + \cos(3+n) \left(-\frac{\pi}{2}\right) \right] - \frac{1}{2(3\pi - n\pi)}$$

$$\left[ -\cos(3-n) \frac{\pi}{2} + \cos(3-n) \left(-\frac{\pi}{2}\right) \right]$$

$$= \frac{1}{2(3\pi + n\pi)} \left[ -\cos(3+n) \frac{\pi}{2} + \cos(3+n) \frac{\pi}{2} \right] - \frac{1}{2(3\pi - n\pi)}$$

$$\left[ -\cos(3-n) \frac{\pi}{2} + \cos(3-n) \frac{\pi}{2} \right]$$

$$= 0$$

The Fourier series from the above function is

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L}$$

$$= \frac{\frac{-2}{3\pi}}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{1} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{1}$$

$$= \frac{-1}{3\pi} + \sum_{n=1}^{\infty} a_n \cos n\pi t + \sum_{n=1}^{\infty} 0 \sin n\pi t$$

$$= \frac{-1}{3\pi} + \sum_{n=1}^{\infty} \left\{ \frac{1}{(3\pi+n\pi)} \sin(3+n)\pi/2 + \frac{1}{(3\pi+n\pi)} \sin(3-n)\pi/2 \right\} \cos n\pi t$$

$$= \frac{-1}{3\pi} + \sum_{n=1}^{\infty} \frac{1}{(3\pi+n\pi)} \sin(3+n)\pi/2 + \frac{1}{(3\pi-n\pi)} \sin(3-n)\pi/2 \cos n\pi t$$

(-An)

### Example - 27

Given that,

$$y = f(t) = 1 - t^2 \text{ ----- (1)}$$

is to be represented by a Fourier series expression over the finite interval  $0 < t < 1$ .

Here,

$$T = 2L = 1$$

$$\therefore L = 1/2$$

Now,

$$\begin{aligned} a_0 &= \frac{1}{L} \int_{-L}^L f(t) dt \\ &= \frac{1}{1/2} \int_0^{2L} f(t) dt \\ &= 2 \int_0^{2 \times \frac{1}{2}} f(t) dt \\ &= 2 \int_0^1 f(t) dt \end{aligned}$$

$$= 2 \int_0^1 (1 - t^2) dt$$

$$= 2 \int_0^1 (1) dt - 2 \int_0^1 t^2 dt$$

$$= 2 \int_0^1 dt - 2 \int_0^1 t^2 dt$$

$$= 2 [t]_0^1 - 2 \left[ \frac{t^3}{3} \right]_0^1$$

$$= 2 [1 - 0] - 2 \left[ \frac{1}{3} - 0 \right]$$

$$= 2 \times 1 - 2 \times \frac{1}{3}$$

$$= 2 - \frac{2}{3}$$

$$= \frac{6-2}{3}$$

$$= \frac{4}{3}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{1/2} \int_0^{2L} f(t) \cos \frac{n\pi t}{1/2} dt$$

$$= 2 \int_0^{2 \times \frac{1}{2}} f(t) \cos(2n\pi t) dt$$

$$= 2 \int_0^1 f(t) \cos(2n\pi t) dt$$

$$= 2 \int_0^1 (1-t^2) \cos(2n\pi t) dt$$

$$= 2 \int_0^1 \cos(2n\pi t) dt - 2 \int_0^1 t^2 \cos(2n\pi t) dt$$

Now,

$$\int \cos(2n\pi t) dt = \frac{1}{2n\pi} \sin(2n\pi t) \left[ \int \cos mx dx = \frac{1}{m} \sin mx \right];$$

$$\int \sin mx dx = -\frac{1}{m} \cos mx$$

And,

$$\int t^2 \cos(2n\pi t) dt = t^2 \int \cos(2n\pi t) dt - \int \left\{ (t^2) \frac{d}{dt} \int \cos(2n\pi t) dt \right\} dt$$

$$[\because \int u dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx]$$

$$= t^2 \frac{1}{2n\pi} \sin(2n\pi t) - \int 2t \frac{1}{2n\pi} \sin(2n\pi t) dt$$

$$= \frac{t^2}{2n\pi} \sin(2n\pi t) - \frac{1}{n\pi} \int t \sin(2n\pi t) dt$$

$$= \frac{t^2}{2n\pi} \sin(2n\pi t) - \frac{1}{n\pi} \left[ t \int \sin(2n\pi t) dt - \int \left\{ \frac{d}{dt}(t) \int \sin(2n\pi t) dt \right\} dt \right]$$

$$= \frac{t^2}{2n\pi} \sin(2n\pi t) - \frac{1}{n\pi} \left[ t \frac{1}{2n\pi} (-\cos 2n\pi t) - \int \frac{1}{2n\pi} (-\cos 2n\pi t) dt \right]$$

$$= \frac{t^2}{2n\pi} \sin(2n\pi t) - \frac{1}{n\pi} \left[ -\frac{t}{2n\pi} \cos 2n\pi t + \frac{1}{2n\pi} \int (\cos 2n\pi t) dt \right]$$

$$= \frac{t^2}{2n\pi} \sin(2n\pi t) - \frac{1}{n\pi} \left[ -\frac{t}{2n\pi} \cos 2n\pi t + \frac{1}{2n\pi} \frac{1}{2n\pi} \sin(2n\pi t) \right]$$

$$= \frac{t^2}{2n\pi} \sin(2n\pi t) + \frac{t}{2n^2\pi^2} - \frac{1}{4n^3\pi^3} \sin(2n\pi t) \quad \dots \textcircled{ii}$$

Again,

$$a_n = 2 \int_0^1 \cos(2n\pi t) dt - 2 \int_0^1 t^2 \cos(2n\pi t) dt$$

$$= 2 \left[ \frac{1}{2n\pi} \sin(2n\pi t) \right]_0^1 - 2 \left[ \frac{t^2}{2n\pi} \sin(2n\pi t) + \frac{t}{2n^2\pi^2} \cos 2n\pi t - \frac{1}{4n^3\pi^3} \sin(2n\pi t) \right]_0^1$$

$$= 2 \left[ \frac{1}{2n\pi} \sin(2n\pi) - 0 \right] - 2 \left[ \frac{1}{2n\pi} \sin(2n\pi) - 0 + \right.$$

$$\left. - \frac{1}{2n^2\pi^2} \cos 2n\pi - 0 - \frac{1}{4n^3\pi^3} \sin(2n\pi) + 0 \right]$$

$$= -\frac{1}{n^2\pi^2} \cos(2n\pi) + 0$$

$$= -\frac{1}{n^2\pi^2} \cos(2n\pi)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{L} \int_0^{2L} f(t) \sin \frac{n\pi t}{1/2} dt$$



$$= \frac{1}{1/2} \int_0^{2 \times \frac{1}{2}} f(t) \sin \frac{n\pi t}{1/2} dt$$

$$= 2 \int_0^1 f(t) \sin(2n\pi t) dt$$

$$= 2 \int_0^1 (1-t^2) \sin(2n\pi t) dt$$

$$= 2 \int_0^1 \sin(2n\pi t) dt - 2 \int_0^1 t^2 \sin(2n\pi t) dt$$

Now,

$$\int \sin(2n\pi t) dt = -\frac{1}{2n\pi} \cos(2n\pi t)$$

$$= -\frac{1}{2n\pi} \cos(2n\pi t)$$

And,

$$\int t^2 \sin(2n\pi t) dt = t^2 \int \sin(2n\pi t) dt - \int \left\{ \frac{d}{dt} (t^2) \right\} \int \sin(2n\pi t) dt \int dt$$

$$[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \right\} \int v dx \int dx]$$

$$= t^2 \frac{1}{2n\pi} \{-\cos(2n\pi t)\} - \int 2t \frac{1}{2n\pi} \{-\cos(2n\pi t)\} dt$$

$$= -\frac{t^2}{2n\pi} \cos(2n\pi t) + \frac{1}{n\pi} \int t \cos(2n\pi t) dt$$

$$= -\frac{t^2}{2n\pi} \cos(2n\pi t) + \frac{1}{n\pi} \left[ t \int \cos(2n\pi t) dt - \int \left\{ \frac{d}{dt} (t) \right\} \int \cos(2n\pi t) dt \int dt \right]$$

$$= -\frac{t^2}{2n\pi} \cos(2n\pi t) + \frac{1}{n\pi} \left[ t \frac{1}{2n\pi} \sin(2n\pi t) - \int 1 \frac{1}{2n\pi} \sin(2n\pi t) dt \right]$$

$$= -\frac{t^2}{2n\pi} \cos(2n\pi t) + \frac{1}{n\pi} \left[ \frac{t}{2n\pi} \sin 2n\pi t - \frac{1}{2n\pi} \int \sin(2n\pi t) dt \right]$$

$$= -\frac{t^2}{2n\pi} \cos(2n\pi t) + \frac{1}{n\pi} \left[ \frac{t}{2n\pi} (2n\pi t) - \frac{1}{2n\pi} \frac{1}{2n\pi} (-\cos 2n\pi t) \right]$$

$$= -\frac{t^2}{2n\pi} \cos(2n\pi t) + \frac{t}{2n\pi^2} \sin 2n\pi t + \frac{1}{4n^3\pi^3} \cos(2n\pi t) \quad \text{--- (iii)}$$

$$b_n = 2 \int_0^1 \sin(2n\pi t) dt - 2 \int_0^1 t \sin(2n\pi t) dt$$

$$= -\frac{2}{2n\pi} [\cos(2n\pi t)]_0^1 - 2 \left[ -\frac{t^2}{2n\pi} \cos(2n\pi t) + \frac{t}{2n\pi^2} \sin 2n\pi t + \frac{1}{4n^3\pi^3} \cos(2n\pi t) \right]_0^1$$

$$= -\frac{2}{2n\pi} [\cos(2n\pi) - 1] - 2 \left[ -\frac{1}{2n\pi} \cos(2n\pi) + \frac{1}{2n^2\pi^2} \sin 2n\pi + \frac{1}{4n^3\pi^3} \cos(2n\pi) + 0 - 0 - \frac{1}{4n^3\pi^3} \right]$$

$$= -\frac{1}{n\pi} \cos(2n\pi) + \frac{1}{n\pi} + \frac{1}{n\pi} \cos(2n\pi) - \frac{1}{2n^3\pi^3}$$

$$= \cos(2n\pi) + \frac{4}{2n^3\pi^3}$$

$$= \frac{1}{n\pi} - \frac{1}{2n^3\pi^3} \cos(2n\pi) + \frac{1}{2n^3\pi^3}$$

$$= \frac{1}{n\pi} - \frac{1}{2n^3\pi^3} [\cos(2n\pi) - 1]$$

The Fourier series for the above function is -

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi t}{L}\right)$$

$$= \frac{4/3}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{1/2}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi t}{1/2}\right)$$

$$= \frac{2}{3} + \sum_{n=1}^{\infty} a_n \cos(2n\pi t) + \sum_{n=1}^{\infty} b_n \sin(2n\pi t)$$

$$= \frac{2}{3} + \sum_{n=1}^{\infty} \left[ -\frac{1}{n^3\pi^2} \cos(2n\pi) \right] \cos(2n\pi t) + \sum_{n=1}^{\infty}$$

$$\left[ \frac{1}{n\pi} - \frac{1}{2n^3\pi^3} \{ \cos 2n\pi - 1 \} \right] \sin(2n\pi t)$$

$$= \frac{2}{3} + \sum_{n=1}^{\infty} \left[ -\frac{1}{n^3\pi^2} \cos(2n\pi) \right] \cos(2n\pi t) +$$

$$\sum_{n=1}^{\infty} \left[ \frac{1}{n\pi} - \frac{1}{2n^3\pi^3} \{ \cos 2n\pi - 1 \} \right] \sin(2n\pi t)$$

(Ans)

### Example - 28

$$y = f(t) = 3t; \quad 0 < t < 1$$
$$= 3; \quad 1 < t < 2$$

$$f(t+2) = f(t) \quad \text{Here, } T = 2L = 2 \therefore L = 1$$

Soln:

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{L} \int_0^{2L} f(t) dt$$

$$= \frac{1}{1} \int_0^2 f(t) dt$$

$$= \int_0^2 f(t) dt$$

$$= \int_0^1 f(t) dt + \int_1^2 f(t) dt$$

$$= \int_0^1 3t dt + \int_1^2 3 dt$$

$$= 3 \left[ \frac{t^2}{2} \right]_0^1 + [3t]_1^2$$

$$= \frac{3}{2} [1-0] + 3[2-1]$$

$$= -\frac{3}{2} [1-0] + 3 \times 1$$

$$= \frac{3}{2} + 3$$

$$= \frac{3+0.6}{2}$$

$$= \frac{3}{2}$$

Again,

$$a_n = \frac{1}{L} \int_0^L f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{L} \int_0^{2L} f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{4} \int_0^2 f(t) \cos \frac{n\pi t}{4} dt$$

$$= \int_0^2 f(t) \cos n\pi t dt$$

$$= \int_0^1 f(t) \cos n\pi t dt + \int_1^2 f(t) \cos n\pi t dt$$

$$= \int_0^1 3t \cos n\pi t dt + \int_1^2 \cos n\pi t dt$$

Now,

$$\int 3t \cos n\pi t dt = 3t \int \cos n\pi t dt - \int \left[ \frac{d}{dt}(3t) \int \cos n\pi t dt \right] dt$$

$$[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx]$$

$$= 3t \times \frac{1}{n\pi} \sin n\pi t - \int 3 \times \frac{1}{n\pi} \sin n\pi t dt$$

$$= \frac{3t}{n\pi} \sin n\pi t - \frac{3}{n\pi} \int \sin n\pi t dt$$

$$= \frac{3t}{n\pi} \sin n\pi t - \frac{3}{n\pi} \times \frac{1}{n\pi} (-\cos n\pi t) \left[ \int \cos mx dx = \right.$$

$$\left. \frac{1}{m} \sin mx ; \int \sin mx dx = -\frac{1}{m} \cos mx \right]$$

$$= \frac{3t}{n\pi} \sin n\pi t + \frac{3}{n^2\pi^2} \cos(n\pi t) \quad \text{--- (ii)}$$

And,

$$\int 3 \cos n\pi t dt = 3 \times \frac{1}{n\pi} \sin n\pi t \left[ \int \cos mx dx = \frac{1}{m} \sin mx ; \right.$$

$$\left. \int \sin mx dx = -\frac{1}{m} \cos mx \right]$$

$$= \frac{3}{n\pi} \sin n\pi t \quad \text{--- (iii)}$$

Again,

$$a_n = \int_0^1 3t \cos n\pi t dt + \int_1^2 3 \cos n\pi t dt$$

$$= \left[ \frac{3t}{n\pi} \sin n\pi t + \frac{3}{n^2\pi^2} \cos(n\pi t) \right]_0^1 + \left[ \frac{3}{n\pi} \sin n\pi t \right]_1^2$$

$$= \left[ \frac{3 \times 1}{n\pi} \sin n\pi + \frac{3}{n^2\pi^2} \cos n\pi - \frac{3 \times 0}{n\pi} \sin 0 - \frac{3}{n^2\pi^2} \cos 0 \right] +$$

$$\left[ \frac{3}{n\pi} \sin 2n\pi - \frac{3}{n\pi} \sin n\pi \right]$$

$$= \frac{3}{n^2\pi^2} \cos n\pi - \frac{3}{n^2\pi^2} \times 1$$

$$= \frac{3}{n^2\pi^2} (\cos n\pi - 1)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{L} \int_0^{2L} f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{1} \int_0^2 f(t) \sin \frac{n\pi t}{1} dt$$

$$= \int_0^2 f(t) \sin n\pi t dt$$

$$= \int_0^1 f(t) \sin n\pi t dt + \int_1^2 f(t) \sin n\pi t dt$$

$$= \int_0^1 3t \sin n\pi t dt + \int_1^2 3 \sin n\pi t dt$$

Now,

$$\begin{aligned} \int 3t \sin n\pi t dt &= 3t \int \sin n\pi t - \int \left\{ \frac{d}{dt} (3t) \right\} \int \sin n\pi t dt dt \\ & \quad \left[ \because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \right\} v dx dx \right] \\ &= 3t \times \frac{1}{n\pi} (-\cos n\pi t) - \int 3(-\cos n\pi t) \times \frac{1}{n\pi} dt \\ &= -\frac{3t}{n\pi} \cos n\pi t + \frac{1}{n\pi} \int 3 \cos n\pi t dt \\ &= -\frac{3t}{n\pi} \cos n\pi t + \frac{3}{n\pi} \times \frac{1}{n\pi} \sin n\pi t \\ &= -\frac{3t}{n\pi} \cos n\pi t + \frac{3}{n^2 \pi^2} \sin n\pi t \quad \text{--- (iv)} \end{aligned}$$

And,

$$\begin{aligned} \int 3 \sin n\pi t dt &= 3 \times \frac{1}{n\pi} (-\cos n\pi t) \\ &= -\frac{3}{n\pi} \cos n\pi t \quad \text{--- (v)} \end{aligned}$$

Again,

$$\begin{aligned} b_n &= \int_0^1 3t \sin n\pi t dt + \int_1^2 3 \sin n\pi t dt \\ &= \left[ -\frac{3t}{n\pi} \cos n\pi t + \frac{3}{n^2 \pi^2} \sin n\pi t \right]_0^1 + \left[ -\frac{3}{n\pi} \cos n\pi t \right]_1^2 \\ &= \left[ \frac{-3 \times 1}{n\pi} \cos n\pi + \frac{3}{n^2 \pi^2} \sin n\pi + \frac{3 \times 0}{n\pi} \cos 0 - \frac{3}{n^2 \pi^2} \sin 0 \right] + \\ & \quad \left[ -\frac{3}{n\pi} \cos 2n\pi + \frac{3}{n\pi} \cos n\pi \right] \end{aligned}$$

$$= -\frac{3}{n\pi} \cos n\pi - \frac{3}{n\pi} \cos 2n\pi + \frac{3}{n\pi} \cos n\pi$$

$$= -\frac{3}{n\pi} \cos 2n\pi$$

The Fourier series for the above function is -

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L}$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{1} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{1}$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi t + \sum_{n=1}^{\infty} b_n \sin n\pi t$$

$$= \frac{3/2}{2} + \sum_{n=1}^{\infty} \left( \frac{3}{n^2\pi^2} (\cos n\pi - 1) \cos n\pi t + \sum_{n=1}^{\infty} \left( -\frac{3}{n\pi} \cos 2n\pi \right) \sin n\pi t \right)$$

$$= \frac{3}{4} + \sum_{n=1}^{\infty} \left( \frac{3}{n^2\pi^2} (\cos n\pi - 1) \cos n\pi t + \sum_{n=1}^{\infty} \left( -\frac{3}{n\pi} \cos 2n\pi \right) \sin n\pi t \right)$$

$$= \frac{3}{4} + \sum_{n=1}^{\infty} \left( \frac{3}{n^2\pi^2} (\cos n\pi - 1) \cos n\pi t + \sum_{n=1}^{\infty} \left( -\frac{3}{n\pi} \cos 2n\pi \right) \sin n\pi t \right)$$

(Ans)



### Example-29

$$y = f(t) = t^2 + t \quad ; \quad -\pi < t < \pi$$

$$f(t) = f(t + 2\pi) \quad T = 2\pi = 2L \quad \therefore L = \pi$$

Sol<sup>n</sup>

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (t^2 + t) dt$$

$$= \frac{1}{\pi} \left[ \frac{t^3}{3} + \frac{t^2}{2} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[ \frac{\pi^3}{3} + \frac{\pi^2}{2} - \frac{(-\pi)^3}{3} - \frac{(-\pi)^2}{2} \right]$$

$$= \frac{1}{\pi} \left[ \frac{\pi^3}{3} + \frac{\pi^2}{2} + \frac{\pi^3}{3} - \frac{\pi^2}{2} \right]$$

$$= \frac{1}{\pi} \left[ \frac{2\pi^3}{3} \right]$$

$$= \frac{2\pi^2}{3}$$

Again,

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (t^2 + 1) \cos nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \cos nt dt + \frac{1}{\pi} \int_{-\pi}^{\pi} \cos nt dt$$

Now,

$$\int t^2 \cos nt dt = t^2 \int \cos nt dt - \int \left\{ \frac{d}{dt} (t^2) \int \cos nt dt \right\} dt$$

$$[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx]$$

$$= t^2 \frac{1}{n} \sin nt - \int 2t \frac{1}{n} \sin nt dt$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \int t \sin nt dt$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ t \int \sin nt dt - \int \left\{ \frac{d}{dt} (t) \int \sin nt dt \right\} dt \right]$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ t \frac{1}{n} (-\cos nt) - \int 1 \cdot \frac{1}{n} (-\cos nt) dt \right]$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ t \frac{1}{n} (-\cos nt) + \int 1 \cdot \frac{1}{n} (\cos nt) dt \right]$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ t \frac{1}{n} (-\cos nt) + \frac{1}{n} \int \cos nt dt \right]$$

$$= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ t \cdot \frac{1}{n} (-\cos nt) + \frac{1}{n} \cdot \frac{1}{n} \sin nt \right]$$

$$= \frac{t^2}{n} \sin nt + \frac{2t}{n^2} \cos nt - \frac{2}{n^3} \sin nt \quad \text{--- (iii)}$$

And,

$$\int t \cos nt dt = t \int \cos nt dt - \int \left\{ \frac{d}{dt}(t) \int \cos nt dt \right\} dt$$

$$[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx]$$

$$= t \cdot \frac{1}{n} \sin nt - \int 1 \cdot \frac{1}{n} \sin nt dt$$

$$= \frac{t}{n} \sin nt - \frac{1}{n} \int \sin nt dt$$

$$= \frac{t}{n} \sin nt - \frac{1}{n} \cdot \frac{1}{n} (-\cos nt) \quad \left[ \int \cos mx dx = \frac{1}{m} \sin mx; \right. \\ \left. \int \sin mx dx = -\frac{1}{m} \cos mx \right]$$

$$= \frac{t}{n} \sin nt + \frac{1}{n^2} \cos nt \quad \text{--- (iii)}$$

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \cos nt dt + \frac{1}{\pi} \int_{-\pi}^{\pi} t \cos nt dt$$

$$= \frac{1}{\pi} \left[ \frac{t^2}{n} \sin nt + \frac{2t}{n^2} \cos nt - \frac{2}{n^3} \sin nt \right]_{-\pi}^{\pi} +$$

$$+ \frac{1}{\pi} \left[ \frac{t}{n} \sin nt + \frac{1}{n^2} \cos nt \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[ \frac{\pi^2}{n} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi - \frac{(-\pi)^2}{n} \sin(-n\pi) \right. \\ \left. + \frac{2(-\pi)}{n^2} \cos(-n\pi) - \left(-\frac{2}{n^3}\right) \sin(-n\pi) \right] +$$

$$\begin{aligned}
 & \frac{1}{\pi} \left[ \frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \frac{(-\pi)}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] \\
 &= \frac{1}{\pi} \left[ \frac{\pi^2}{n} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi + \frac{\pi^2}{n} \sin n\pi + \right. \\
 & \quad \left. \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi \right] + \frac{1}{\pi} \\
 & \quad \left[ \frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \frac{\pi}{n} \sin n\pi - \frac{1}{n^2} \cos n\pi \right] \\
 &= \frac{1}{\pi} \left[ 0 + \frac{2\pi}{n^2} \cos n\pi - 0 + 0 + \frac{2\pi}{n^2} \cos n\pi - 0 \right] + \frac{1}{\pi} \\
 & \quad \left[ 0 + \frac{1}{n^2} \cos n\pi - 0 - \frac{1}{n^2} \cos n\pi \right] \\
 &= \frac{1}{\pi} \left[ \frac{4\pi}{n^2} \cos n\pi \right] + \frac{1}{\pi} \left[ \frac{1}{n^2} \cos n\pi - \frac{1}{n^2} \cos n\pi \right] \\
 &= \frac{4\pi}{\pi n^2} \cos n\pi + 0 \\
 &= \frac{4}{n^2} \cos n\pi
 \end{aligned}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (t^2 + t) \sin nt dt$$



$$= \frac{1}{n} \int_{-\pi}^{\pi} t^2 \sin nt dt + \frac{1}{n} \int_{-\pi}^{\pi} t \sin nt dt$$

Now,

$$\int t^2 \sin nt dt = t^2 \int \sin nt dt - \int \left\{ \frac{d}{dt} (t^2) \int \sin nt dt \right\} dt$$

$$[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx]$$

$$= t^2 \int \sin nt dt - \int 2t \cdot \frac{1}{n} (-\cos nt) dt$$

$$= t^2 \frac{1}{n} (-\cos nt) + \frac{2}{n} \int t \cos nt dt$$

$$= t^2 \frac{1}{n} (-\cos nt) + \frac{2}{n} \left[ t \int \cos nt dt - \int \left\{ \frac{d}{dt} (t) \int \cos nt dt \right\} dt \right]$$

$$= t^2 \frac{1}{n} (-\cos nt) + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin t - \int 1 \cdot \frac{1}{n} \sin nt dt \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin t - \frac{1}{n} \int \sin nt dt \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin nt + \frac{1}{n} \frac{1}{n} (-\cos nt) \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin nt + \frac{1}{n^2} \cos nt \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2t}{n^2} \sin nt + \frac{2}{n^3} \cos nt \dots \text{--- (iv)}$$



Again,

$$\begin{aligned} \int t \sin nt \, dt &= t \int \sin nt \, dt - \int \left\{ \frac{d}{dt} (t) \int \sin nt \, dt \right\} dt \\ & \quad \left[ \because \int u \, dv = u \int v \, dx - \int \left\{ \frac{d}{dx} (u) \int v \, dx \right\} dx \right] \\ &= t \int \sin nt \, dt - \int 1 \cdot \frac{1}{n} (-\cos nt) \, dt \\ &= t \cdot \frac{1}{n} (-\cos nt) + \frac{1}{n} \int \cos nt \, dt \\ &= t \cdot \frac{1}{n} (-\cos nt) + \frac{1}{n} \cdot \frac{1}{n} \sin nt \left[ \int \cos mx \, dx = \frac{1}{m} \sin mx; \int \sin mx \, dx = -\frac{1}{m} \cos mx \right] \end{aligned}$$

$$= -t \cdot \frac{1}{n} \cos nt + \frac{1}{n^2} \sin nt \quad \text{--- (v)}$$

$$\begin{aligned} \therefore b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \sin nt \, dt + \frac{1}{\pi} \int_{-\pi}^{\pi} t \sin nt \, dt \\ &= \frac{1}{\pi} \left[ -t^2 \cdot \frac{1}{n} \cos nt + \frac{2t}{n^2} \sin nt + \frac{2}{n^3} \cos nt \right]_{-\pi}^{\pi} + \\ & \quad \frac{1}{\pi} \left[ -t \cdot \frac{1}{n} \cos nt + \frac{1}{n^2} \sin nt \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cos n\pi + \frac{2\pi}{n^2} \sin n\pi + \frac{2}{n^3} \cos n\pi - \frac{-(-\pi)^2}{2} \cos(-n\pi) - \frac{2(-\pi)}{n^2} \sin(-n\pi) - \frac{2}{n^3} (\cos(-n\pi)) \right] + \\ & \quad \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{-(-\pi)}{n} \cos(-n\pi) - \frac{1}{n^2} (-\sin(-n\pi)) \right] \\ &= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cos n\pi + \frac{2\pi}{n^2} \sin n\pi + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi - \right. \\ & \quad \left. \frac{2\pi}{n^2} \sin n\pi - \frac{2}{n^3} \cos n\pi \right] + \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \right. \end{aligned}$$

$$\begin{aligned}
& \left[ \frac{\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin n\pi \right] \\
&= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cos n\pi + 0 + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi - 0 - \frac{2}{n^3} \cos n\pi \right] \\
&+ \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi + 0 - \frac{\pi}{n} \cos n\pi + 0 \right] \\
&= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cos n\pi + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi - \frac{2}{n^3} \cos n\pi \right] + \\
&\quad \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi - \frac{\pi}{n} \cos n\pi \right] \\
&= \frac{1}{\pi} \left[ -\frac{\pi}{n} \cos n\pi - \frac{\pi}{n} \cos n\pi \right] \\
&= -\frac{2}{n} \cos n\pi
\end{aligned}$$

The Fourier series for the above function is

$$\begin{aligned}
f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L} \\
&= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{\pi} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{\pi} \\
&= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n t + \sum_{n=1}^{\infty} b_n \sin n t \\
&= \frac{\frac{2\pi^3}{3}}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos n t + \sum_{n=1}^{\infty} \left( -\frac{2}{n} \cos n\pi \right) \sin n t \\
&= \frac{2\pi^3/3}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos n t + \sum_{n=1}^{\infty} \left( -\frac{2}{n} \cos n\pi \right) \sin n t
\end{aligned}$$

(Ans)

### Example-30

$$y = f(t) = t; \quad 0 \leq t \leq \pi$$

$$= \pi; \quad \pi \leq t < 2\pi$$

Here,  $T = 2L = 2\pi \quad \therefore L = \pi$

(a) Sketch the function for 3 cycles.

(b) Find the Fourier series for the function.

(a)

Soln:

$$f(t) = \begin{cases} t & 0 \leq t < \pi \\ \pi & \pi \leq t < 2\pi \end{cases}$$

$$f(t) = \begin{cases} t & 0 \leq t < \pi \\ \pi & \pi \leq t < 2\pi \end{cases}$$

$$f(t) = \begin{cases} t & 0 \leq t < \pi \\ \pi & \pi \leq t < 2\pi \end{cases}$$

$$f(t) = \begin{cases} t & 0 \leq t < \pi \\ \pi & \pi \leq t < 2\pi \end{cases}$$

$$f(t) = \begin{cases} t & 0 \leq t < \pi \\ \pi & \pi \leq t < 2\pi \end{cases}$$

$$f(t) = \begin{cases} t & 0 \leq t < \pi \\ \pi & \pi \leq t < 2\pi \end{cases}$$

(b)

Soln: We have the fourier series is

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi t}{L}\right) \quad \text{--- (1)}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_0^{2\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_0^{\pi} f(t) dt + \frac{1}{\pi} \int_{\pi}^{2\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_0^{\pi} t dt + \frac{1}{\pi} \int_{\pi}^{2\pi} \pi dt$$

$$= \frac{1}{\pi} \int_0^{\pi} t dt + \int_{\pi}^{2\pi} dt$$

$$= \frac{1}{\pi} \left[ \frac{t^2}{2} \right]_0^{\pi} + [t]_{\pi}^{2\pi}$$

$$= \frac{1}{2\pi} [\pi^2 - 0] + [2\pi - \pi]$$

$$= \frac{1}{2\pi} \times \pi^2 + \pi$$

$$= \frac{\pi}{2} + \pi$$

$$= \frac{3\pi}{2}$$

We have,

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L f(t) \cos\left(n \frac{\pi}{L} t\right) dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos\left(n \frac{\pi}{\pi} t\right) dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt \\ &= \frac{1}{\pi} \int_0^{\pi} f(t) \cos nt dt + \frac{1}{\pi} \int_{\pi}^{2\pi} f(t) \cos nt dt \\ &= \frac{1}{\pi} \int_0^{\pi} t \cos nt dt + \frac{1}{\pi} \int_{\pi}^{2\pi} \pi \cos nt dt \\ &= \frac{1}{\pi} \int_0^{\pi} t \cos nt dt + \int_{\pi}^{2\pi} \cos nt dt \quad \text{--- (iii)} \end{aligned}$$



Now,

$$\text{let, } I = \int t \times \cos nt \, dt$$

$$= t \int \cos nt \, dt - \int \left[ \frac{d}{dt}(t) \times \int \cos nt \, dt \right] dt$$

$$= t \times \left( \frac{\sin nt}{n} \right) - \int 1 \times \left( \frac{\sin nt}{n} \right) dt$$

$$= \frac{t}{n} \sin nt - \frac{1}{n} \times \left( -\frac{\cos nt}{n} \right)$$

$$= \frac{t}{n} \sin nt + \frac{1}{n^2} \cos nt \quad \text{--- (iv)}$$

Putting Result of  $\int t \times \cos nt \, dt$  in (iii)

$$a_n = \frac{1}{\pi} \int_0^{\pi} t \times \cos nt \, dt + \int_{\pi}^{2\pi} \cos nt \, dt$$

$$= \frac{1}{\pi} \left[ \frac{t}{n} \sin nt + \frac{1}{n^2} \cos nt \right]_0^{\pi} + \left[ \frac{\sin nt}{n} \right]_{\pi}^{2\pi}$$

$$= \frac{1}{\pi} \left[ \frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \frac{0}{n} \sin(n \times 0) - \frac{1}{n^2} \cos(n \times 0) \right] +$$

$$\frac{1}{n^2} [\sin(n \times 2\pi) - \sin n\pi]$$

$$= \frac{1}{\pi} \left[ \frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \frac{0}{n} \sin(n \times 0) - \frac{1}{n^2} \cos 0 \right] +$$

$$\frac{1}{n^2} [\sin 2n\pi - \sin n\pi]$$

$$= \frac{1}{\pi} \left( \frac{\pi}{n} \times 0 + \frac{1}{n^2} \cos n\pi - 0 - \frac{1}{n^2} \times 1 \right) + \frac{1}{n^2} \sin n\pi$$

$$= \frac{1}{\pi} \left( \frac{1}{n^2} \cos n\pi - \frac{1}{n^2} \right) + 0$$



$$= \frac{1}{n\pi} (\cos n\pi - 1) \left[ \frac{1}{n} \right]$$

Again,

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin\left(n \frac{\pi}{L} t\right) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin\left(n \frac{\pi}{\pi} t\right) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_0^{2\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_0^{\pi} f(t) \sin nt dt + \frac{1}{\pi} \int_{\pi}^{2\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_0^{\pi} t \times \sin nt dt + \frac{1}{\pi} \int_{\pi}^{2\pi} \pi \times \sin nt dt$$

$$= \frac{1}{\pi} \int_0^{\pi} t \times \sin nt dt + \int_{\pi}^{2\pi} \sin nt dt \quad \text{--- (vi)}$$

Now,

$$\text{let } I = \int t \times \sin nt dt.$$

$$= t \int \sin nt dt - \int \left\{ \frac{d}{dt} (t) \int \sin nt dt \right\} dt$$

$$= t \left[ \frac{-\cos nt}{n} \right] - \int 1 \times \left[ \frac{-\cos nt}{n} \right] dt$$

$$= -\frac{t}{n} \cos nt + \frac{1}{n} \int \cos nt dt$$

$$= -\frac{t}{n} \cos nt + \frac{1}{n} \left[ \frac{\sin nt}{n} \right] \quad (1)$$

$$= -\frac{1}{n} \cos nt + \frac{1}{n} \sin nt$$

$$= \frac{1}{n^2} \sin nt - \frac{t}{n} \cos nt \dots\dots (vii)$$

Putting result of  $\int -x \sin t dt$  in (vi)

$$b_n = \frac{1}{\pi} \int_0^{\pi} t \sin nt dt + \int_{\frac{\pi}{2}}^{2\pi} \sin nt dt$$

$$= \frac{1}{\pi} \left[ \frac{1}{n^2} \sin nt - \frac{1}{n} \cos nt \right]_0^{\pi} + \left[ \frac{-\cos nt}{n} \right]_{\pi}^{2\pi}$$

$$= \frac{1}{\pi} \left( \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos n\pi - \frac{1}{n^2} \sin(n\pi) - \frac{0}{n} \cos(n\pi) \right)$$

$$= \frac{1}{n} (\cos 2\pi - \cos n\pi)$$

$$= \frac{1}{\pi} \left( \frac{1}{n^2} \times 0 - \frac{\pi}{n} \cos n\pi - \frac{1}{n^2} \sin 0 - \frac{0}{n} \cos b \right) - \frac{1}{n} (\cos 2n\pi - \cos n\pi)$$

$$= \frac{1}{\pi} \left( 0 - \frac{\pi}{n} \cos n\pi - 0 - 0 \right) - \frac{1}{n} (\cos 2n\pi - \cos n\pi)$$

$$= \frac{1}{\pi} \left( -\frac{\pi}{n} \cos n\pi \right) - \frac{1}{n} (\cos 2n\pi - \cos n\pi)$$

$$= -\frac{1}{n} \cos n\pi - \frac{1}{n} (\cos 2n\pi - \cos n\pi)$$

$$= -\frac{1}{n} \cos 2n\pi - \frac{1}{n} \cos 2n\pi + \frac{1}{n} \cos n\pi$$

$$= -\frac{1}{n} \cos 2n\pi \quad \dots \dots \dots \text{viii}$$

Putting the values of  $a_0, a_n, b_n$  in (ii)

The Fourier series for the above function is:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi t}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi t}{L}\right)$$

$$= \frac{3\pi/2}{2} + \sum_{n=1}^{\infty} \frac{1}{\pi n^2} (\cos n\pi - 1) \cos\left(\frac{n\pi t}{\pi}\right) + \sum_{n=1}^{\infty} -\frac{1}{n}$$

$$\cos 2n\pi \sin\left(\frac{n\pi t}{\pi}\right)$$

$$= \frac{3\pi}{4} + \sum_{n=1}^{\infty} \frac{1}{\pi n^2} (\cos n\pi - 1) \cos nt - \sum_{n=1}^{\infty} \frac{1}{n} \cos 2n\pi \sin nt$$

$$= \frac{3\pi}{4} + \sum_{n=1}^{\infty} \frac{1}{\pi n^2} (\cos n\pi - 1) \cos nt - \sum_{n=1}^{\infty} \frac{1}{n} \cos 2n\pi \sin nt$$

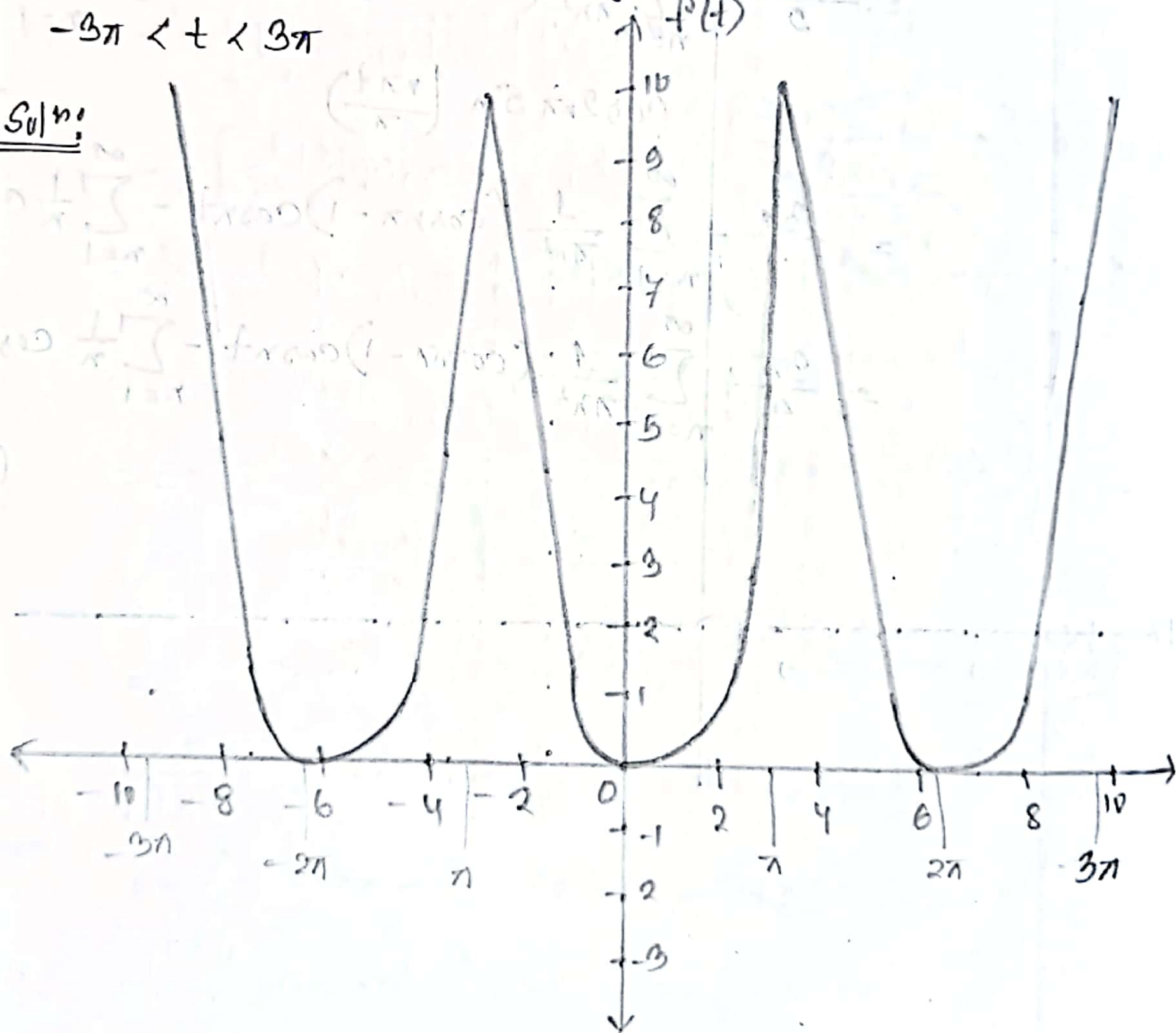
(Am)

### Example - 3.1

• If  $y = f(t) = t^2$  over the interval  $-\pi < t < \pi$  and has period  $2\pi$ . Hence,  $T = 2L = 2\pi \therefore L = \pi$

① Sketch three cycles of  $y = f(t)$  in the interval  $-3\pi < t < 3\pi$

Soln:



(b) Find the fourier series of the function

Sol<sup>n</sup>: Let,

$$y = f(t) = t^2 \quad \text{----- (i)}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 dt$$

$$= \frac{1}{\pi} \left[ \frac{t^3}{3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[ \frac{\pi^3}{3} - \frac{(-\pi)^3}{3} \right]$$

$$= \frac{1}{\pi} \left[ \frac{\pi^3}{3} + \frac{\pi^3}{3} \right]$$

$$= \frac{1}{\pi} \left[ \frac{2\pi^3}{3} \right]$$

$$a_0 = \frac{2\pi^2}{3}$$

Again,

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos nt dt$$

$$= \frac{1}{n} \int_{-\pi}^{\pi} t^2 \cos nt \, dt$$

Now,

$$\begin{aligned} \int t^2 \cos nt \, dt &= t^2 \int \cos nt \, dt - \int \left\{ \frac{d}{dt} (t^2) \int \cos nt \, dt \right\} dt \\ &\quad \left[ \because \int u \, dv = u \int v \, dx - \int \left\{ \frac{d}{dx} (u) \int v \, dx \right\} dx \right] \\ &= \frac{t^2}{n} \sin nt - \int \left\{ 2t \sin nt \frac{\sin nt}{n} \right\} dt \\ &= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ t \int \sin nt - \int \left\{ \frac{d}{dt} (t) \int \sin nt \, dt \right\} dt \right] \\ &= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ t \cdot \frac{(-\cos nt)}{n} - \int \frac{(-\cos nt)}{n} dt \right] \\ &= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ \frac{-t \cos nt}{n} + \int \frac{\cos nt}{n} dt \right] \\ &= \frac{t^2}{n} \sin nt - \frac{2}{n} \left[ \frac{-t \cos nt}{n} + \frac{\sin nt}{n} \right] \\ &= \frac{t^2}{n} \sin nt + \frac{2t}{n^2} \cos nt - \frac{2}{n^3} \sin nt \end{aligned}$$

$$\int t^2 \cos nt \, dt = \frac{t^2}{n} \sin nt + \frac{2t}{n^2} \cos nt - \frac{2}{n^3} \sin nt \quad \text{--- (ii)}$$

Again,

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \cos nt \, dt \\ &= \frac{1}{\pi} \left[ \frac{t^2}{n} \sin nt + \frac{2t}{n^2} \cos nt - \frac{2}{n^3} \sin nt \right]_{-\pi}^{\pi} \end{aligned}$$

$$= \frac{1}{\pi} \left[ \frac{\pi^2}{n} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi - \frac{(-\pi)^2}{n} \sin n(-\pi) - \frac{2(-\pi)}{n^2} \cos(-\pi) + \frac{2}{n^3} \sin n(-\pi) \right]$$

$$= \frac{1}{\pi} \left[ \frac{\pi^2}{n} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi + \frac{2\pi}{n^2} \cos n\pi - \frac{2}{n^3} \sin n\pi \right]$$

$$[\because \sin(-\theta) = -\sin \theta, \cos(-\theta) = \cos \theta]$$

$$= \frac{1}{\pi} \left[ 0 + \frac{2\pi}{n^2} \cos n\pi - 0 + 0 + \frac{2\pi}{n^2} \cos n\pi - 0 \right]$$

$$= \frac{1}{\pi} \left[ \frac{2\pi}{n^2} \cos n\pi + \frac{2\pi}{n^2} \cos n\pi \right]$$

$$= \frac{1}{\pi} \left[ \frac{4\pi}{n^2} \cos n\pi \right]$$

$$= \frac{4}{n^2} \cos n\pi$$

Again,

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t^2 \sin nt dt$$

(iii)

Now,

$$\int t^2 \sin nt \, dt = t^2 \int \sin nt \, dt - \int \left\{ \frac{d}{dt} (t^2) \int \sin nt \, dt \right\} dt$$

$$\left[ \because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int \frac{d}{dx} (v) \right. \right. \\ \left. \left. \int v dx \right\} dx \right]$$

$$= t^2 \int \sin nt \, dt - \int \left\{ 2t \cdot \frac{1}{n} (-\cos nt) \right\} dt$$

$$= t^2 \frac{1}{n} (-\cos nt) + \frac{2}{n} \int t \cos nt \, dt$$

$$= t^2 \frac{1}{n} (-\cos nt) + \frac{2}{n} \left[ t \int \cos nt + \int \left\{ \frac{d}{dt} (t) \right. \right. \\ \left. \left. \int \cos nt \, dt \right\} dt \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin nt - \int 1 \cdot \frac{1}{n} \right. \\ \left. \sin nt \, dt \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin nt - \frac{1}{n} \int \sin nt \, dt \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin nt - \frac{1}{n} \cdot \frac{1}{n} \right. \\ \left. (-\cos nt) \right]$$

$$= -t^2 \frac{1}{n} \cos nt + \frac{2}{n} \left[ t \cdot \frac{1}{n} \sin nt + \frac{1}{n^2} \right. \\ \left. \cos nt \right]$$

$$\int t^2 \sin nt \, dt = -t^2 \frac{1}{n} \cos nt + \frac{2t}{n^2} \sin nt + \frac{2}{n^3} \cos nt$$

--- (iii)



Again,

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} -t^2 \sin nt \, dt$$

$$= \frac{1}{\pi} \left[ -t^2 \frac{1}{n} \cos nt + \frac{2t}{n^2} \sin nt + \frac{2}{n^3} \cos nt \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cos n\pi + \frac{2\pi}{n^2} \sin n\pi + \frac{2}{n^3} \cos n\pi - \frac{-(-\pi)^2}{n} \right]$$

$$\cos(-\pi) - \frac{2(-\pi)}{n^2} \sin(-\pi) - \frac{2}{n^3} \cos n(-\pi) \Big]$$

$$= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cos n\pi + \frac{2\pi}{n^2} \sin n\pi + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi - \frac{2\pi}{n^2} \sin n\pi - \frac{2}{n^3} \cos n\pi \right]$$

$$[\sin(-\theta) = -\sin \theta ; \cos(-\theta) = \cos \theta]$$

$$= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cos n\pi + 0 + \frac{2}{n^3} \cos n\pi + \frac{\pi^2}{n} \cos n\pi - 0 - \frac{2}{n^3} \cos n\pi \right]$$

$$= \frac{1}{\pi} \left[ -\frac{\pi^2}{n} \cancel{\cos n\pi} + \frac{2}{n^3} \cancel{\cos n\pi} + \frac{\pi^2}{n} \cancel{\cos n\pi} - \frac{2}{n^3} \cancel{\cos n\pi} \right]$$

$$= \frac{1}{\pi} [0]$$

$$= 0$$

The Fourier series for the above function is,

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi t + \sum_{n=1}^{\infty} b_n \sin n\pi t$$

$$= \frac{2\pi^2/3}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos n\pi t + 0$$

$$= \frac{2\pi^2/3}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos n\pi t$$

$$= \frac{2\pi^2/3}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos n\pi \cos n\pi t$$

(Am)

Home task no 2

Fourier Transformation

- Representation of Aperiodic Signals. The extension of a Fourier series for a non periodic function is known as the Fourier transforms.

Q: Find the Fourier transformation of the function

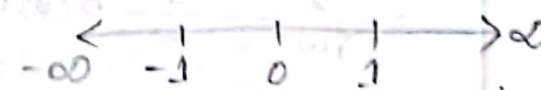
$$f(t) = \begin{cases} 1, & 0 \leq t < 1 \\ -1, & -1 \leq t < 0 \\ 0, & 1+t > 1 \end{cases}$$

Sol<sup>n</sup>:-

Given that,

$$f(t) = \begin{cases} 1, & 0 \leq t < 1 \\ -1, & -1 \leq t < 0 \\ 0, & 1+t > 1 \end{cases}$$

We know that,

$$\begin{aligned} g(\omega) &= \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt \\ &= \int_{-1}^0 f(t) \cdot e^{-i\omega t} dt + \int_0^1 f(t) \cdot e^{-i\omega t} dt + \int_1^{\infty} f(t) \cdot e^{-i\omega t} dt \end{aligned}$$


$$= \int_{-\infty}^{-1} 0 \cdot e^{-i\omega t} dt + \int_{-1}^0 -1 \cdot e^{-i\omega t} dt + \int_0^1 1 \cdot e^{-i\omega t} dt + \int_1^{\infty} 0 \cdot e^{-i\omega t} dt$$

$$= - \int_{-1}^0 e^{-i\omega t} dt + \int_0^1 e^{-i\omega t} dt$$

$$= \left[ \frac{e^{-i\omega t}}{-i\omega} \right]_{-1}^0 + \left[ \frac{e^{-i\omega t}}{-i\omega} \right]_0^1$$

$$= \left[ \frac{e^{-i\omega \cdot 0}}{i\omega} - \frac{e^{-i\omega(-1)}}{-i\omega} \right] + \left[ \frac{e^{-i\omega \cdot 1}}{-i\omega} - \frac{e^{-i\omega \cdot 0}}{-i\omega} \right]$$

$$= \frac{1}{i\omega} - \frac{e^{-i\omega}}{i\omega} - \frac{e^{-i\omega}}{i\omega} + \frac{1}{i\omega}$$

$$= \frac{2}{i\omega} - \frac{e^{i\omega}}{i\omega} - \frac{e^{-i\omega}}{i\omega}$$

$$= \frac{2}{i\omega} - \frac{1}{i\omega} (e^{i\omega} + e^{-i\omega})$$

$$= \frac{2}{i\omega} - \frac{1}{\omega} 2 \cos \omega \left[ \frac{e^{i0} + e^{-i0}}{2} = \cos 0 \right]$$

$$= \frac{2}{i\omega} (1 - \cos \omega)$$

(Ans)

### Example - 42 (Home task)

• Find Fourier Transform of  $f(t) = e^{-|t|}$   
or,

Find Fourier Transform of

$$f(t) = e^{-t} \quad ; \quad t > 0$$
$$= e^t \quad ; \quad t < 0$$

Sol<sup>n</sup>: Given that,

$$f(t) = e^{-t} \quad ; \quad t > 0$$
$$= e^t \quad ; \quad t < 0$$

(i)

We know that,

$$g(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$
$$= \int_{-\infty}^0 f(t) e^{-i\omega t} dt + \int_0^{\infty} f(t) e^{-i\omega t} dt$$
$$= \int_{-\infty}^0 e^t e^{-i\omega t} dt + \int_0^{\infty} e^{-t} e^{-i\omega t} dt$$
$$= \int_{-\infty}^0 e^t e^{-i\omega t} dt + \int_0^{\infty} e^t e^{-i\omega t} dt$$

$$= \int_{-\infty}^0 e^{t-i\omega t} dt + \int_0^{\infty} e^{-t-i\omega t} dt$$

$$= \int_{-\infty}^0 e^{(1-i\omega)t} dt + \int_0^{\infty} e^{-(1+i\omega)t} dt$$

$$= \left[ \frac{e^{(1-i\omega)t}}{(1-i\omega)} \right]_{-\infty}^0 + \left[ \frac{e^{-(1+i\omega)t}}{-(1+i\omega)} \right]_0^{\infty}$$

$$= \left[ \frac{e^{(1-i\omega) \cdot 0}}{(1-i\omega)} - \frac{e^{(1-i\omega)(-\infty)}}{(1-i\omega)} \right] + \left[ \frac{e^{-(1+i\omega)\infty}}{-(1+i\omega)} - \frac{e^{-(1+i\omega) \cdot 0}}{-(1+i\omega)} \right]$$

$$= \left[ \frac{e^0}{1-i\omega} - \frac{e^{-\infty}}{(1-i\omega)} \right] + \left[ \frac{e^{-\infty}}{-(1+i\omega)} - \frac{e^{-0}}{-(1+i\omega)} \right]$$

$$= \left[ \frac{1}{(1-i\omega)} - \frac{1}{e^{\infty}(1-i\omega)} \right] + \left[ \frac{e^{-\infty}}{-(1+i\omega)} - \frac{e^{-0}}{-(1+i\omega)} \right]$$

$$= \frac{1}{(1-i\omega)} \left[ 1 - \frac{1}{e^{\infty}} \right] + \frac{-1}{1+i\omega} \left[ \frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$= \frac{1}{1-i\omega} \left[ 1 - \frac{1}{\infty} \right] + \frac{-1}{1+i\omega} \left[ \frac{1}{\infty} - \frac{1}{1} \right]$$

$$= \frac{1}{1-i\omega} [1-0] + \frac{-1}{1+i\omega} \left[ \frac{1}{\infty} - \frac{1}{1} \right]$$

$$= \frac{1}{1-i\omega} + \frac{-1}{1+i\omega} [0-1]$$

$$= \frac{1}{1-i\omega} + \frac{1}{1+i\omega}$$



$$= \frac{1-i\omega + 1-i\omega}{(1-i\omega)(1+i\omega)}$$

$$= \frac{2}{(1-i\omega)(1+i\omega)}$$

$$= \frac{2}{1-i^2\omega^2}$$

$$= \frac{2}{1+\omega^2} [i^2 = -1]$$

Example-44 (Home task)

Find the Fourier Transform of

$$f(t) = 1 + \frac{t}{a} \quad ; -a < t < 0$$

$$= 1 - \frac{t}{a} \quad ; 0 < t < a$$

$$= 0 \quad ; \text{otherwise}$$

Soln: Given that,

$$f(t) = 1 + \frac{t}{a} \quad ; -a < t < 0$$

$$= 1 - \frac{t}{a} \quad ; 0 < t < a$$

$$= 0 \quad ; \text{otherwise}$$

We have,

$$f(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [g(w)] e^{i\omega t} d\omega$$

We know that,

$$g(w) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^{-a} f(t) e^{-i\omega t} dt + \int_{-a}^0 f(t) e^{-i\omega t} dt + \int_0^a f(t) e^{-i\omega t} dt +$$

$$\int_a^{\infty} f(t) e^{-i\omega t} dt$$

$$= \int_{-\infty}^{-a} 0 \cdot e^{-i\omega t} dt + \int_{-a}^0 \left(1 + \frac{t}{a}\right) \cdot e^{-i\omega t} dt + \int_0^a \left(1 - \frac{t}{a}\right) e^{-i\omega t} dt$$

$$+ \int_a^{\infty} 0 \cdot e^{-i\omega t} dt$$

$$= \int_{-a}^0 \left(1 + \frac{t}{a}\right) e^{-i\omega t} dt + \int_0^a \left(1 - \frac{t}{a}\right) \cdot e^{-i\omega t} dt$$

$$= \int_{-a}^0 1 \cdot e^{-i\omega t} dt + \int_{-a}^0 \frac{t}{a} \cdot e^{-i\omega t} dt + \int_0^a 1 \cdot e^{-i\omega t} dt$$

$$+ \int_0^a \left(-\frac{t}{a}\right) \cdot e^{-i\omega t} dt$$

$$= \int_{-a}^0 e^{-i\omega t} dt + \frac{1}{a} \int_{-a}^0 t \cdot e^{-i\omega t} dt + \int_0^a e^{-i\omega t} dt - \frac{1}{a} \int_0^a t \cdot e^{-i\omega t} dt$$

$$= \int_{-a}^0 e^{-i\omega t} dt + \int_0^a e^{-i\omega t} dt + \frac{1}{a} \int_{-a}^0 t \cdot e^{-i\omega t} dt - \frac{1}{a} \int_0^a t \cdot e^{-i\omega t} dt \quad \text{--- (i)}$$

Now,

$$\int t \cdot e^{-i\omega t} dt = t \int e^{-i\omega t} dt - \int \left\{ \frac{d}{dt}(t) \int e^{-i\omega t} dt \right\} dt$$

$$[\int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx]$$

$$= t \left[ \frac{e^{-i\omega t}}{-i\omega} \right] - \int \left\{ 1 \cdot \left[ \frac{e^{-i\omega t}}{-i\omega} \right] \right\} dt$$

$$= t \left[ \frac{e^{-i\omega t}}{-i\omega} \right] + \int \frac{e^{-i\omega t}}{i\omega} dt$$

$$= t \left[ \frac{e^{-i\omega t}}{-i\omega} \right] + \frac{1}{i\omega} \int e^{-i\omega t} dt$$

$$= t \left[ \frac{e^{-i\omega t}}{-i\omega} \right] + \frac{1}{i\omega} \left[ \frac{e^{-i\omega t}}{-i\omega} \right]$$

$$= t \left[ \frac{e^{-i\omega t}}{-i\omega} \right] - \frac{1}{i^2 \omega^2} e^{-i\omega t}$$

$$= t \left[ \frac{e^{-i\omega t}}{-i\omega} \right] - \frac{1}{-\omega^2} e^{-i\omega t} \quad [ \because i^2 = -1 ]$$

$$= t \left[ \frac{e^{-i\omega t}}{-i\omega} \right] + \frac{1}{\omega^2} e^{-i\omega t}$$

$$= \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \quad \text{--- (ii)}$$

Putting the result of  $\int t \cdot e^{-i\omega t} dt$  in (i) we get,

$$\begin{aligned}
 g(\omega) &= \int_{-a}^0 e^{-i\omega t} dt + \int_0^a e^{-i\omega t} dt + \frac{1}{a} \int_{-a}^0 t \cdot e^{-i\omega t} dt - \frac{1}{a} \int_0^a t \cdot e^{-i\omega t} dt \\
 &= \left[ \frac{e^{-i\omega t}}{-i\omega} \right]_{-a}^0 + \left[ \frac{e^{-i\omega t}}{-i\omega} \right]_0^a + \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_{-a}^0 \\
 &\quad - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
 &= \left[ \frac{e^{-i\omega \cdot 0}}{-i\omega} - \frac{e^{-i\omega(-a)}}{-i\omega} \right] + \left[ \frac{e^{-i\omega \cdot a}}{-i\omega} - \frac{e^{-i\omega \cdot 0}}{-i\omega} \right] + \frac{1}{a} \\
 &\quad \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_{-a}^0 - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
 &= \left[ \frac{e^{-0}}{-i\omega} - \frac{e^{i\omega a}}{-i\omega} \right] + \left[ \frac{e^{-i\omega \cdot a}}{-i\omega} - \frac{e^{-0}}{-i\omega} \right] + \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_{-a}^0 \\
 &\quad - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
 &= \left[ \frac{1}{-i\omega} - \frac{e^{i\omega a}}{-i\omega} \right] + \left[ \frac{e^{-i\omega \cdot a}}{-i\omega} - \frac{1}{-i\omega} \right] + \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_{-a}^0 \\
 &\quad - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
 &= \left[ \frac{1}{-i\omega} + \frac{e^{i\omega a}}{i\omega} \right] + \left[ \frac{e^{-i\omega \cdot a}}{-i\omega} + \frac{1}{i\omega} \right] + \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_{-a}^0 \\
 &\quad - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
 &= \left[ -\frac{1}{i\omega} + \frac{e^{i\omega a}}{i\omega} \right] + \left[ -\frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{i\omega} \right] + \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_{-a}^0 \\
 &\quad - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a
 \end{aligned}$$

$$\begin{aligned}
& e^{-i\omega t} \Big|_{-a}^0 - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
&= -\frac{1}{i\omega} + \frac{e^{ai\omega}}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{i\omega} + \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
&= \frac{e^{ai\omega}}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
&= \frac{e^{ai\omega}}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{-i\omega} \cdot 0 \cdot e^{-i\omega \cdot 0} + \frac{1}{\omega^2} e^{-i\omega \cdot 0} - \frac{1}{-i\omega} (-a) e^{-i\omega(-a)} - \frac{1}{\omega^2} e^{-i\omega(-a)} \right] - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
&= \frac{e^{ai\omega}}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{a} \left[ 0 + \frac{1}{\omega^2} e^{-0} - \frac{1}{-i\omega} (-a) e^{i\omega a} - \frac{1}{\omega^2} e^{i\omega a} \right] - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
&= \frac{e^{ai\omega}}{i\omega} - \frac{e^{-i\omega a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{\omega^2} \cdot 1 + \frac{1}{-i\omega} a e^{i\omega a} - \frac{1}{\omega^2} e^{i\omega a} \right] - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a \\
&= \frac{e^{ai\omega}}{i\omega} - \frac{e^{-i\omega a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{\omega^2} - \frac{1}{i\omega} a e^{i\omega a} - \frac{1}{\omega^2} e^{i\omega a} \right] - \frac{1}{a} \left[ \frac{1}{-i\omega} t e^{-i\omega t} + \frac{1}{\omega^2} e^{-i\omega t} \right]_0^a
\end{aligned}$$

$$= \frac{e^{i\omega} a}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{\omega^2} - \frac{1}{i\omega} a e^{i\omega a} - \frac{1}{\omega^2} e^{i\omega a} \right] -$$

$$\frac{1}{a} \left[ -\frac{1}{i\omega} a e^{-i\omega a} + \frac{1}{\omega^2} e^{-i\omega a} - \frac{1}{-i\omega} \cdot 0 \cdot e^{-i\omega \cdot 0} - \frac{1}{\omega^2} e^{-i\omega \cdot 0} \right]$$

$$= \frac{e^{i\omega} a}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{\omega^2} - \frac{1}{i\omega} a e^{i\omega a} - \frac{1}{\omega^2} e^{i\omega a} \right] -$$

$$\frac{1}{a} \left[ -\frac{1}{i\omega} a e^{-i\omega a} + \frac{1}{\omega^2} e^{-i\omega a} - \frac{1}{\omega^2} e^{-0} \right]$$

$$= \frac{e^{i\omega} a}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{\omega^2} - \frac{1}{i\omega} a e^{i\omega a} - \frac{1}{\omega^2} e^{i\omega a} \right] -$$

$$\frac{1}{a} \left[ -\frac{1}{i\omega} a e^{-i\omega a} + \frac{1}{\omega^2} e^{-i\omega a} - \frac{1}{\omega^2} \cdot 1 \right]$$

$$= \frac{e^{i\omega} a}{i\omega} - \frac{e^{-i\omega \cdot a}}{i\omega} + \frac{1}{a} \left[ \frac{1}{\omega^2} - \frac{1}{i\omega} a e^{i\omega a} - \frac{1}{\omega^2} e^{i\omega a} \right] -$$

$$\frac{1}{a} \left[ -\frac{1}{i\omega} a e^{-i\omega a} + \frac{1}{\omega^2} e^{-i\omega a} - \frac{1}{\omega^2} \right]$$

$$= \frac{e^{i\omega} a}{i\omega} - e \frac{e^{-i\omega \cdot a}}{i\omega} + \left[ \frac{1}{a\omega^2} - \frac{a e^{i\omega a}}{a i\omega} - \frac{e^{i\omega a}}{a\omega^2} \right] -$$

$$\left[ \frac{-a e^{-i\omega a}}{a i\omega} + \frac{e^{-i\omega a}}{a\omega^2} - \frac{1}{a\omega^2} \right]$$

$$= \frac{e^{i\omega} a}{i\omega} - \frac{e^{-i\omega a}}{i\omega} + \frac{1}{a\omega^2} - \frac{a e^{i\omega a}}{a i\omega} - \frac{e^{i\omega a}}{a\omega^2} +$$

$$\frac{a e^{-i\omega a}}{a i\omega} - \frac{e^{-i\omega a}}{a\omega^2} + \frac{1}{a\omega^2}$$

$$= \frac{1}{i\omega} (e^{i\omega} a - e^{-i\omega \cdot a}) + \frac{1}{a\omega^2} + \frac{1}{a\omega^2} + \frac{a e^{-i\omega a}}{a i\omega} -$$

$$\frac{a e^{i\omega a}}{a i\omega} - \frac{e^{i\omega a}}{a\omega^2} - \frac{e^{-i\omega \cdot a}}{a\omega^2}$$

$$= \frac{1}{i\omega} (e^{ai\omega} - e^{-i\omega a}) + \frac{2}{a\omega^2} + \frac{ae^{-i\omega a}}{ai\omega} - \frac{ae^{i\omega a}}{ai\omega} -$$

$$= \frac{1}{i\omega} (e^{ai\omega} - e^{-i\omega a}) + \frac{2}{a\omega^2} + \frac{e^{i\omega a}}{a\omega^2} - \frac{e^{-i\omega a}}{a\omega^2} + \frac{e^{-i\omega a}}{i\omega} - \frac{e^{i\omega a}}{i\omega} -$$

$$= \frac{e^{i\omega a}}{i\omega} - \frac{e^{-i\omega a}}{i\omega} + \frac{2}{a\omega^2} + \frac{e^{-i\omega a}}{i\omega} - \frac{e^{i\omega a}}{i\omega} -$$

$$= \frac{2}{a\omega^2} - \frac{e^{i\omega a}}{a\omega^2} - \frac{e^{-i\omega a}}{a\omega^2}$$

$$= \frac{1}{a\omega^2} (2 - e^{i\omega a} - e^{-i\omega a})$$

(Am)

## Laplace transformation

$$\text{Formula: } \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) \cdot e^{-st} dt$$

↑ এই নিয়মের মাধ্যমে আসবে

Q: Find the Laplace transformation of the function  $f(t) = 1$

Soln: Given that,

$$f(t) = 1$$

We know that,

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) \cdot e^{-st} dt$$

$$\text{or, } \mathcal{L}\{1\} = \int_0^{\infty} 1 \cdot e^{-st} dt$$

$$= \int_0^{\infty} e^{-st} dt$$

$$= \left[ \frac{e^{-st}}{-s} \right]_0^{\infty}$$

$$= \frac{e^{-s \cdot \infty}}{-s} - \frac{e^{-s \cdot 0}}{-s}$$

$$= \frac{e^{-\infty}}{-s} + \frac{e^0}{s}$$

$$= \frac{1}{s} \left( 1 - \frac{1}{e^{\infty}} \right)$$

$$= \frac{1}{s} \left( 1 - \frac{1}{\infty} \right)$$

$$= \frac{1}{s} (1 - 0)$$

$$= \frac{1}{s} \underline{\underline{(A_m)}}$$

### Example - 56 (Home task)

• Find Laplace Transform of  $f(t) = a$ .

Sol<sup>n</sup>: We know that,

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) \cdot e^{-st} dt$$

Given that,

$$f(t) = a$$

$$\text{or, } \mathcal{L}\{f(t)\} = \int_0^{\infty} a \cdot e^{-st} dt$$

$$= a \int_0^{\infty} e^{-st} dt$$

$$= a \left[ \frac{e^{-st}}{-s} \right]_0^{\infty}$$

$$= -\frac{a}{s} \left[ \frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$= -\frac{a}{s} [e^{-\infty} - e^{-0}]$$

$$= -\frac{a}{s} \left[ \frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$= -\frac{a}{s} \left[ \frac{1}{\infty} - \frac{1}{1} \right]$$

$$= -\frac{1}{s} [0 - 1]$$

$$= \frac{1}{s} \quad (\text{Ans})$$

### Example - 57 (Home task)

- Find the Laplace Transform of  $f(t) = t$ .

Soln:

Given that,

$$f(t) = t$$

We know that,

$$L\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

$$\text{or, } L(t) = \int_0^{\infty} t e^{-st} dt \quad \text{--- (i)}$$

Now,

$$\int t \cdot e^{-st} dt = t \int e^{-st} dt - \int \left[ \frac{d}{dt} (t) \int e^{-st} dt \right] dt$$

$$\left[ \because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx \right]$$

$$= t \frac{e^{-st}}{-s} - \int 1 \cdot \frac{e^{-st}}{-s} dt \left[ \int e^{-mx} = \frac{e^{-mx}}{-m} \right]$$

$$= t \cdot \frac{e^{-st}}{-s} + \frac{1}{s} \cdot \frac{e^{-st}}{-s}$$

$$= -\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st}$$

From eqn (i)  $\Rightarrow$

$$L(t) = \int_0^{\infty} t e^{-st} dt$$

$$= \left[ -\frac{t}{s} e^{-st} \right]_0^{\infty} + \left[ -\frac{1}{s^2} e^{-st} \right]_0^{\infty}$$

$$= 0 + \left[ -\frac{1}{s^2} e^{-\infty} + \frac{1}{s^2} e^0 \right]$$

$$= 0 + \left( 0 + \frac{1}{s^2} \right) \left[ \begin{array}{l} e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0 \\ e^0 = 1 \end{array} \right]$$

$$= \frac{1}{s^2}$$

### Example - 58 (Home task)

• Find the Laplace Transform of  $f(t) = e^{at}$ .

Sol<sup>n</sup>: Given that,

$$f(t) = e^{at}$$

We know that,

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

$$\therefore \mathcal{L}\{f(e^{at})\} = \int_0^{\infty} e^{at} \cdot e^{-st} dt$$

$$= \int_0^{\infty} e^{at-st} dt$$

$$= \int_0^{\infty} e^{-st+at} dt$$

$$= \int_0^{\infty} e^{-(s-a)t} dt$$

$$= \left[ \frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty}$$

$$= -\frac{1}{(s-a)} [e^{-\infty} - e^{-0}]$$

$$= -\frac{1}{(s-a)} \left[ \frac{1}{e^{\infty}} - \frac{1}{e^0} \right]$$

$$= -\frac{1}{(s-a)} \left[ \frac{1}{\infty} - \frac{1}{1} \right]$$

$$= -\frac{1}{(s-1)} [0-1]$$

$$= \frac{1}{s-1} \quad (\text{Am})$$

Example - 59 (Home task)

Example - 60

• Find Laplace Transform of  $f(t) = \sin at$ .

Soln: Given that,

$$f(t) = \sin at$$

We know that,

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) \cdot e^{-st} \cdot dt$$

$$\text{or, } \mathcal{L}(\sin at) = \int_0^{\infty} \sin at \cdot e^{-st} dt \quad \text{----- (i)}$$

Again,

We know that,

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\therefore e^{iat} = \cos at + i \sin at$$

Here,

$\cos at$  is the real (R) part of  $e^{iat}$  and  $\sin at$  is the imaginary (I) part of  $e^{iat}$

That is  $\cos at = \text{Re}(e^{iat})$  and  $\sin at = \text{Im}(e^{iat})$ .

from (i) We have,

$$\mathcal{L}(\sin at) = \int_0^{\infty} \sin at \cdot e^{-st} dt$$

$$= \int_0^{\infty} \mathcal{I} e^{iat} \cdot e^{-st} dt$$

$$= \int_0^{\infty} \mathcal{I} e^{iat} \cdot e^{-st} dt$$

$$= \mathcal{I} \int_0^{\infty} e^{(ia-s)t} dt$$

$$= \mathcal{I} \int_0^{\infty} e^{-(s-ia)t} dt$$

$$= \mathcal{I} \left\{ \left[ \frac{e^{-(s-ia)t}}{-(s-ia)} \right]_0^{\infty} \right\}$$

$$= \mathcal{I} \left\{ \frac{e^{-(s-ia)\infty}}{-(s-ia)} - \frac{e^{-(s-ia) \cdot 0}}{-(s-ia)} \right\}$$

$$= \mathcal{I} \left\{ 0 - \frac{e^{-0}}{-(s-ia)} \right\}$$

$$= \mathcal{I} \left[ \frac{1}{s-ia} \right]$$

$$= \mathcal{I} \left\{ \frac{s+ia}{(s+ia)(s-ia)} \right\}$$

$$= \mathcal{I} \left\{ \frac{s+ia}{s^2 - i^2 a^2} \right\}$$

$$= -1 \left\{ \frac{s+ia}{s^2+a^2} \right\} [i \equiv -1]$$

$$= -1 \left\{ \frac{a}{s^2+a^2} + i \frac{a}{s^2+a^2} \right\} \text{ --- (ii)}$$

Taking -thet imaginary part from (ii)

$$L(f(t)) = L(\sin at)$$

$$= \frac{a}{s^2+a^2} \quad (Am)$$

Example 64

• Find  $L\{e^{-at}t\} = ?$  / State  $\rightarrow$  fi

Soln:

### Example - 64

- State first shift theorem using the theorem  
Find the value of  $\mathcal{L}\{e^{-at}t\} = ?$

Soln:

Statement: The first shift theorem states that,  
if  $\mathcal{L}\{f(t)\} = f(s)$  then  $\mathcal{L}\{e^{-at}f(t)\} = f(s+a)$

According to the question.

let,

$$f(t) = t$$

Now

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{t\} = \frac{1}{s^2}$$

According to the first shift theorem

$$f(s) = \frac{1}{s^2} \text{ ---- (i)}$$

Also

$$\begin{aligned}\mathcal{L}\{e^{-at}t\} &= f(s+a) \\ &= \frac{1}{(s+a)^2} \text{ [ by (i) ]}\end{aligned}$$

$$\therefore \mathcal{L}\{e^{-at}t\} = f(s+a) = \frac{1}{(s+a)^2} \quad \underline{\underline{Ans}}$$

## Multiplication theorem

Example - 71: Find  $L\{t \cdot t\} = ?$

Soln:-

According to the question -

let  $f(t) = t$

Now  $L\{f(t)\} = L(t) = \frac{1}{s^2}$

According to multiplication theorem

$$f(s) = \frac{1}{s^2} \text{ --- (1)}$$

$$L\{t \cdot f(t)\} = -\frac{d}{ds} [f(s)]$$

$$\text{or, } L\{t \cdot t\} = -\frac{d}{ds} \left( \frac{1}{s^2} \right)$$

$$\text{or, } L\{t \cdot t\} = -\frac{d}{ds} (s^{-2})$$

$$\text{or, } L\{t \cdot t\} = -(-2)s^{-3}$$

$$\therefore L\{t \cdot t\} = \frac{2}{s^3}$$

## Division theorem

Ex-75: Find Laplace Transform of  $\left\{ \frac{\sin 2t}{t} \right\}$

Soln: According to the ques,

$$f(t) = \sin 2t$$

Now,

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \mathcal{L}(\sin 2t) \\ &= \frac{2}{s^2+4} \end{aligned}$$

According to division theorem,

$$f(s) = \frac{2}{s^2+4}$$

$$\text{or, } \mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty f(s) ds$$

$$\begin{aligned} \text{or, } \mathcal{L}\left\{\frac{\sin 2t}{2}\right\} &= \int_s^\infty \frac{2}{s^2+4} ds \\ &= 2 \int_s^\infty \frac{ds}{s^2+4} \end{aligned}$$

$$= 2 \cdot \frac{1}{2} [\tan^{-1} s/2]$$

$$= \tan^{-1} \infty - \tan^{-1} s/2$$

$$= \frac{\pi}{2} - \tan^{-1} s/2 \quad (\underline{Ans})$$

## Define Unit Step-function

14.10.2023

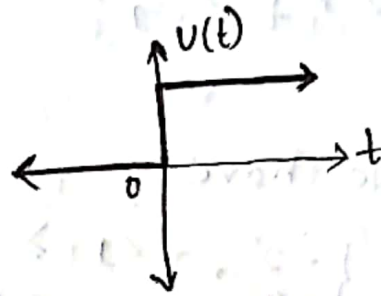
Define Unit Step-function: The unit step-function  $u(t)$

( $x_m$  আছড়ে পারে)

is defined as-

$$f(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Which is called Heaviside's unit-function.



Example-89 ( $x_m$  আছড়ে পারে)

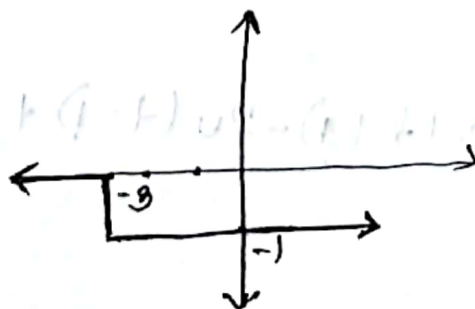
Soln: Given that,

$$x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$$

Now for

$-u(t+3)$  we have

$$\begin{aligned} -u(t+3) &= \begin{cases} 1, & t+3 \geq 0 \\ 0, & t+3 < 0 \end{cases} \\ &= \begin{cases} 1, & t \geq -3 \\ 0, & t < -3 \end{cases} \end{aligned}$$

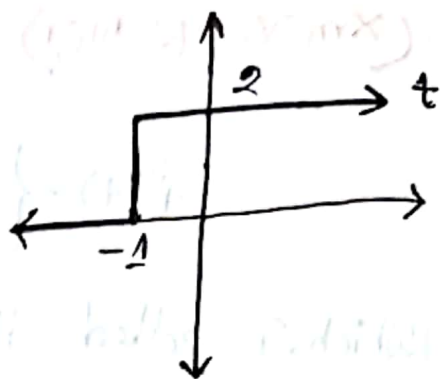


Again for,

$2v(t+1)$  we have

$$2v(t+1) = \begin{cases} 2, & t+1 \geq 0 \\ 0, & t+1 < 0 \end{cases}$$

$$= \begin{cases} 2, & t \geq -1 \\ 0, & t < -1 \end{cases}$$

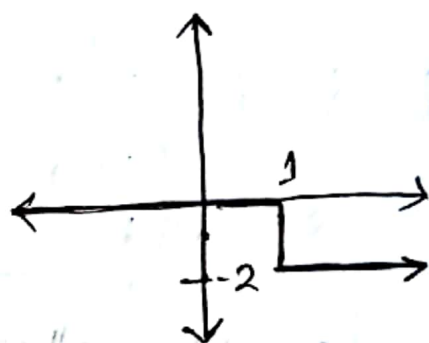


Again for

$-2v(t-1)$  we have

$$-2v(t-1) = \begin{cases} -2, & t-1 \geq 0 \\ 0, & t-1 < 0 \end{cases}$$

$$= \begin{cases} -2, & t \geq 1 \\ 0, & t < 1 \end{cases}$$

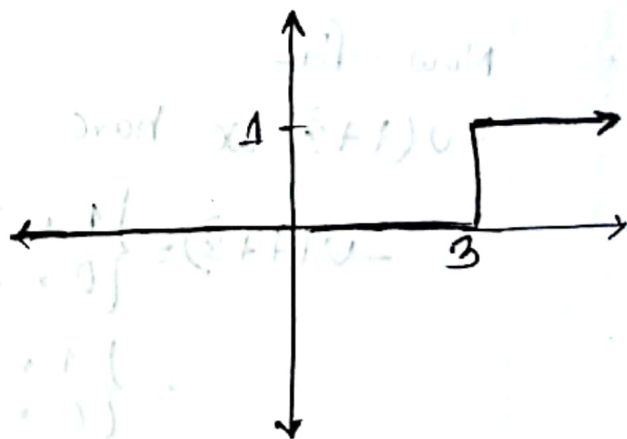


Again for,

$v(t-3)$  we have

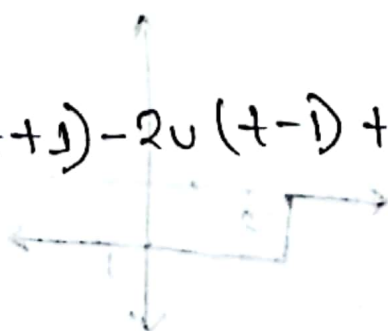
$$v(t-3) = \begin{cases} 1, & t-3 \geq 0 \\ 0, & t-3 < 0 \end{cases}$$

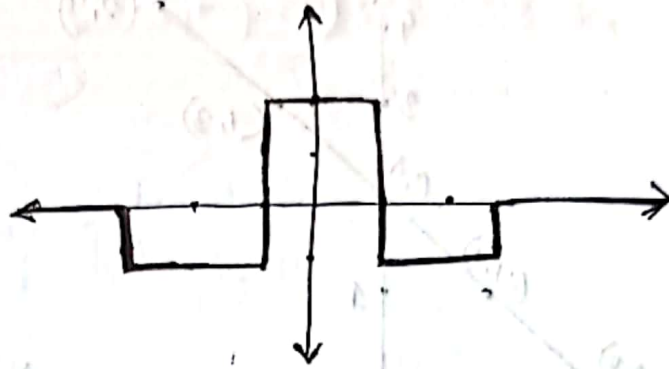
$$= \begin{cases} 1, & t \geq 3 \\ 0, & t < 3 \end{cases}$$



Finally for,

$$x(t) = -v(t+3) + 2v(t+1) - 2v(t-1) + v(t-3)$$





Define Ramp-function: The ramp-function  $r(t)$  is define as -

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



Example : 89/90 যেকোনো একটি গ্রাফকে

Example-90

$$x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$$

Given that,

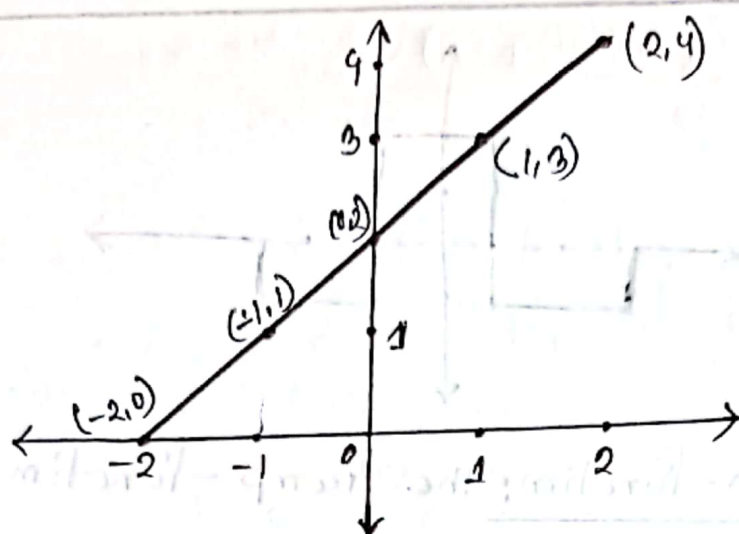
$$x(t) = r(t+2) + r(t+1) - r(t-1) + r(t-2)$$

Now for,

$r(t+2)$  we have,

$$r(t+2) = \begin{cases} t+2, & t \geq -2 \\ 0, & t < -2 \end{cases}$$

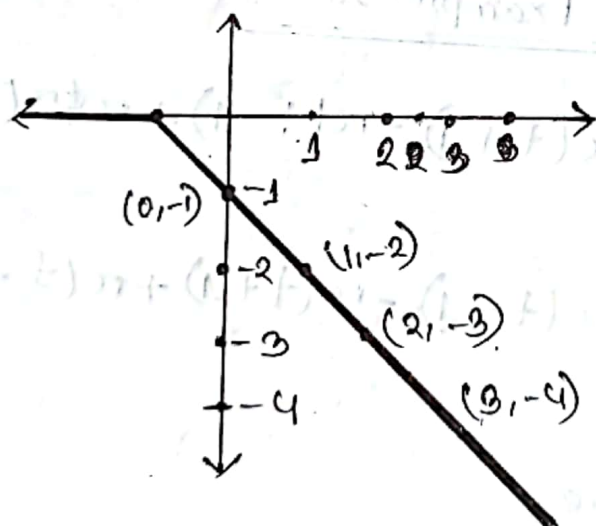
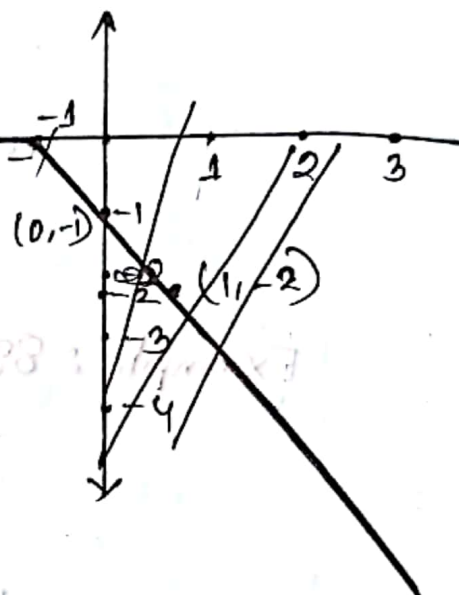
| t              | -2 | -1 | 0 | 1 | 2 |
|----------------|----|----|---|---|---|
| $r(t+2) = t+2$ | 0  | 1  | 2 | 3 | 4 |



Again,

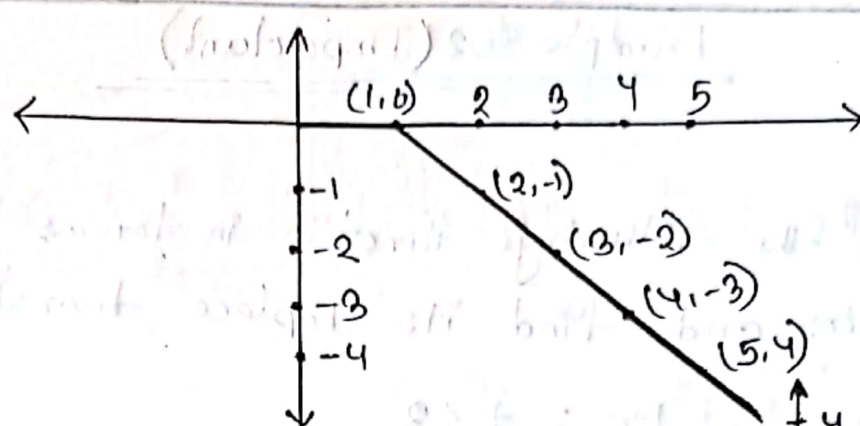
$$-n(t+1) = \begin{cases} -(t+1), & t \geq -1 \\ 0, & t < -1 \end{cases}$$

| $t$                | -1 | 0  | 1  | 2  | 3  |
|--------------------|----|----|----|----|----|
| $-n(t+1) = -(t+1)$ | 0  | -1 | -2 | -3 | -4 |



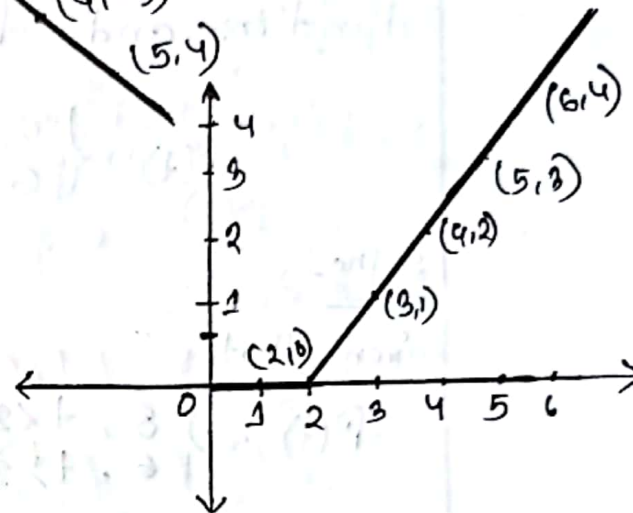
$$-n(t-1) = \begin{cases} -(t-1), & t \geq 1 \\ 0, & t < 1 \end{cases}$$

| $t$                | 1 | 2  | 3  | 4  | 5  |
|--------------------|---|----|----|----|----|
| $-n(t-1) = -(t-1)$ | 0 | -1 | -2 | -3 | -4 |



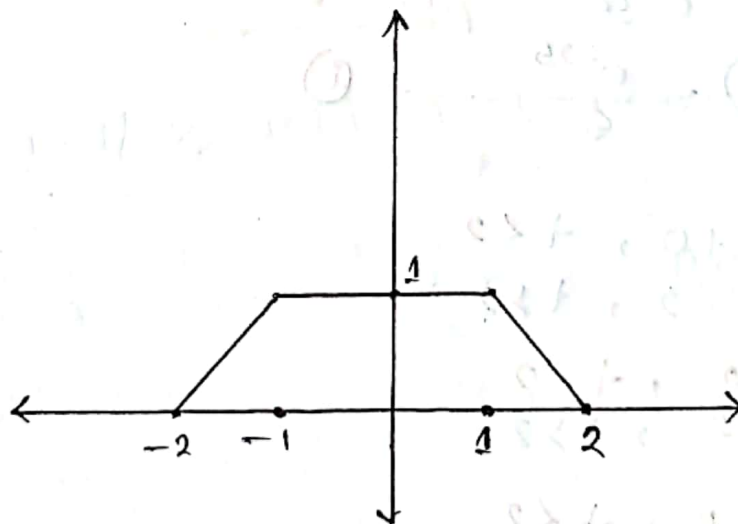
$$r(t-2) = \begin{cases} t-2, & t \geq 2 \\ 0, & t < 2 \end{cases}$$

|                |   |   |   |   |   |              |
|----------------|---|---|---|---|---|--------------|
| $t$            | 2 | 3 | 4 | 5 | 6 | <del>7</del> |
| $r(t-2) = t-2$ | 0 | 1 | 2 | 3 | 4 | <del>5</del> |



Final for

$$x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$$



### Example - 92 (Important)

- Express the following function in terms of unit step functions and find its Laplace transform:

$$f(t) = \begin{cases} 8 & ; t < 2 \\ 6 & ; t > 2 \end{cases}$$

Soln:-

Given that:

$$f(t) = \begin{cases} 8, & t < 2 \\ 6, & t > 2 \end{cases}$$

We have

$$L(u(t-a)) = \frac{e^{-as}}{s}$$

$$\therefore L(u(t-2)) = \frac{e^{-2s}}{s} \quad \text{--- (1)}$$

Now,

$$f(t) = \begin{cases} 8+0, & t < 2 \\ 8-2, & t > 2 \end{cases}$$

$$= 8 + \begin{cases} 0, & t < 2 \\ -2, & t > 2 \end{cases}$$

$$= 8 + \begin{cases} 0, & t < 2 \\ -2, & t > 2 \end{cases}$$

$$= 8 + (-2) \begin{cases} 0, & t < 2 \\ 1, & t > 2 \end{cases}$$

$$= 8 + (-2) \begin{cases} 1, & t > 2 \\ 0, & t < 2 \end{cases}$$

$$= 8 + (-2) \cup (t-2)$$

$$= 8 - 2 \cup (t-2)$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{8 - 2 \cup (t-2)\}$$

$$= \mathcal{L}\{8\} - 2 \mathcal{L}\{ \cup (t-2) \}$$

$$= 8 \mathcal{L}\{1\} - 2 \mathcal{L}\{ \cup (t-2) \}$$

$$= 8 \times \frac{1}{s} - 2 \times \frac{e^{-2s}}{s} \left[ \because \mathcal{L}\{1\} = \frac{1}{s} \right. \\ \left. \mathcal{L}\{ \cup (t-2) \} = \frac{e^{-2s}}{s} \right]$$

Example - 99

17.10.2023

अस सूचक -

$$\textcircled{1} \mathcal{L}^{-1}\left(\frac{1}{s}\right) = 1$$

$$\textcircled{2} \mathcal{L}^{-1}(as) = a$$

$$\textcircled{3} \mathcal{L}^{-1}\left(\frac{1}{s^2}\right) = t$$

$$\textcircled{4} \mathcal{L}^{-1}\left(\frac{1}{s-1}\right) = e^{at}$$

$$\textcircled{5} \mathcal{L}^{-1}\left(\frac{1}{s-2}\right) = e^{2t}$$

$$\textcircled{10} \mathcal{L}^{-1}\left(\frac{a}{s^2 - a^2}\right) = \sinh at$$

$$\textcircled{12} \mathcal{L}^{-1}\left(\frac{s}{s^2 - a^2}\right) = \cosh at$$

$$\textcircled{6} \mathcal{L}^{-1}\left(\frac{1}{s^2 + a^2}\right) = \sin at$$

$$\textcircled{7} \mathcal{L}^{-1}\left(\frac{2}{s^2 + 2^2}\right) = \sin 2t$$

$$\textcircled{8} \mathcal{L}^{-1}\left(\frac{s}{s^2 + a^2}\right) = \cos at$$

$$\textcircled{9} \mathcal{L}^{-1}\left(\frac{2}{s^2 + 2^2}\right) = \cos 2t$$

$$\textcircled{11} \mathcal{L}^{-1}\left(\frac{2}{s^2 - 2^2}\right) = \sinh 2t$$

$$\textcircled{13} \mathcal{L}^{-1}\left(\frac{s}{s^2 - 2^2}\right) = \cosh 2t$$

### Example - 100

• Find the inverse Laplace transforms of the following:

(i)  $\frac{1}{s-2}$

Soln: Given that,

$$f(s) = \frac{1}{s-2}$$

We have,

$$L(f(t)) = L(e^{at}) = \frac{1}{s-a}$$

$$\therefore e^{at} = L^{-1}\left(\frac{1}{s-a}\right)$$

$$\therefore e^{2t} = L^{-1}\left(\frac{1}{s-2}\right)$$

$$\therefore L^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \quad \underline{\underline{(Am)}}$$

(ii)  $L^{-1}\left(\frac{s}{s^2-16}\right) = ?$

Soln: We have,

$$L(f(t)) = L(\cosh at) = \frac{s}{s^2-a^2}$$

$$\cosh at = L^{-1}\left(\frac{s}{s^2-a^2}\right)$$

$$\cosh 4t = L^{-1}\left(\frac{s}{s^2-4^2}\right)$$

$$\therefore L^{-1}\left(\frac{s}{s^2-4^2}\right) = \cosh 4t \quad \underline{\underline{(Am)}}$$

$$(iii) \frac{5}{s^2 + 25}$$

Sol<sup>n</sup>: We have,

$$\mathcal{L}^{-1}(f(s)) = \mathcal{L}^{-1}\left(\frac{1}{s^2 + a^2}\right)$$

$$= \frac{1}{a} \sin at$$

$$\therefore \sin at = \mathcal{L}^{-1}\left(\frac{1}{s^2 + a^2}\right)$$

$$\therefore \mathcal{L}^{-1}\left(\frac{5}{s^2 + 5^2}\right) = \sin 5t$$

$$= \sin 5t$$

### Example - 101

• Find inverse Laplace transform of:  $\frac{s+4}{s(s-1)(s-2)}$

Soln:-

Let,

$$\frac{s+4}{s(s-1)(s-2)} = f(s)$$

$$\frac{s+4}{s(s-1)(s-2)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s-2} \quad \text{--- (1)}$$

$$\text{or, } s+4 = A(s-1)(s-2) + Bs(s-2) + Cs(s-1)$$

Now,

$$\text{Where } s=0 \text{ then } 4=2A$$

$$\text{or, } A=2$$

Again,

$$\text{When } s=1 \text{ then } 5=-B$$

$$\text{or, } B=-5$$

$$\text{Also when } s=2 \text{ then } 6=2C$$

$$\text{or, } C=3$$

Now, Putting the value of A, B, C in eqn (1)  
we get,

$$\frac{s+4}{s(s-1)(s-2)} = \frac{2}{s} + \frac{-5}{s-1} + \frac{3}{s-2}$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{s+4}{s(s-1)(s-2)}\right\} = \mathcal{L}^{-1}\left(\frac{2}{s} + \frac{-5}{s-1} + \frac{3}{s-2}\right)$$

$$= 2h^{-1}\left(\frac{1}{s}\right) - 5h^{-1}\left(\frac{1}{s-1}\right) + 3h^{-1}\left(\frac{1}{s-2}\right) \quad \text{--- (2)}$$

Again,

We know that,

$$h(s) = \frac{1}{s}$$

$$\therefore h^{-1}\left(\frac{1}{s}\right) = 1 \quad \text{--- (3)}$$

Again

$$h(s) = \frac{1}{s-1}$$

$$\therefore h^{-1}\left(\frac{1}{s-1}\right) = e^{at}$$

$$\text{or, } h^{-1}\left(\frac{1}{s-1}\right) = e^t \quad \text{--- (4)}$$

$$\text{Similarly } h^{-1}\left(\frac{1}{s-2}\right) = e^{2t} \quad \text{--- (5)}$$

Putting these values in eqn (2), (4), (5) in (2) we get,

$$\begin{aligned} h^{-1}\left\{\frac{s+4}{s(s-1)(s-2)}\right\} &= 2 \cdot 1 - 5 \cdot e^t + 3e^{2t} \\ &= 2 - 5e^t + 3e^{2t} \quad (Ans) \end{aligned}$$